



Geostatistical assessment of Spatial climate dynamics using mono window machine learning algorithm for decoding land cover and demographic shifts influence on thermal environment

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Received: 30 March 2024 / Accepted: 19 August 2025 / Published online: 30 September 2025
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Abstract

Rapid urbanization and climate change create significant challenges for land cover dynamics and demographic patterns in growing cities. This study employs advanced geospatial methods and machine learning algorithms to examine the complex relationships between seasonal land surface temperature (LST) variations and land use/land cover (LULC) changes in urban and rural areas. Using Landsat satellite imagery from 1998 to 2022, we quantified how urban growth, and environmental changes affect seasonal temperature patterns in Dhaka, Bangladesh. Our analysis reveals dramatic landscape transformations: 98% increase in urban built-up areas, accompanied by substantial losses of vegetation (48%) and water bodies (19%). These changes have intensified urban heat island effects and altered local climate patterns. Rural areas experienced similarly rapid development, with a 120% increase in built-up zones and an estimated 30% decrease in water bodies. Our seasonal analysis demonstrates significant temperature increases during summer months, exceeding 36.8 °C, directly linked to LULC changes, providing deeper insights into land cover-temperature relationships. Geographically Weighted Regression (GWR) analysis revealed that urban development positively correlates with rising temperatures, particularly in densely built areas, while vegetated regions show negative correlations with temperature. These relationships vary spatially, highlighting the importance of accounting for spatial heterogeneity in urban climate studies. The GWR model significantly outperformed ordinary least squares regression, with R^2 improving from 0.578 to 0.901, demonstrating the value of spatially explicit modeling approaches. This research enhances our understanding of urbanization's environmental impacts and provides valuable insights for policymakers and urban planners. The findings support the development of climate-resilient urban and rural environments capable of addressing ongoing global challenges including climate change and rapid urbanization. Our methodology offers a replicable framework for analyzing urban climate dynamics in other rapidly expanding cities worldwide.

Keywords Climate Dynamics · Thermal Environment · Machine Learning · Spatial Heterogeneity · Sustainable Development Strategies · Climate mitigation

1 Introduction

Climate change presents a pressing global concern with far-reaching implications for human livelihoods, environmental sustainability, and ecosystem health. Land use/land cover (LULC) alterations can exacerbate heat wave effects while diminishing ecological integrity in densely populated urban areas and their surrounding regions (Kafy et al. 2021). Rapid

land cover transformation profoundly impacts regional environments and development patterns, with regional characteristics becoming increasingly dependent on land use dynamics (Das et al. 2021). The high heat storage capacity inherent in urban environments creates localized areas of elevated temperature, manifesting as the urban heat island (UHI) effect—a phenomenon that causes temperatures in metropolitan areas to be noticeably higher than in nearby

rural areas (Das et al. 2021; Tran et al. 2017). Land surface temperature (LST) serves as a reliable indicator of Earth's surface energy balance, providing crucial insights into the complex ecological, environmental, and geochemical processes occurring within Earth's systems. Beyond serving as a measurement tool, LST has the potential to influence these complex dynamics directly (Ibrahim 2017). As countries worldwide experience unprecedented urban expansion, with urban areas growing at an alarming pace, the impact of surface temperature on these processes becomes increasingly significant.

According to the United Nations, urban populations are projected to increase from an estimated 55% currently to 68% by 2050 (United Nation 2018). Urban expansion will add 2.5 billion more people to cities by 2050, with 90% of this increase occurring in Asia and Africa (Mukherjee and Singh 2020). Urbanization stands as the primary catalyst driving demographic shifts, social transformations, economic dynamics, and environmental changes within nations (Choudhury et al. 2019; Ding and Shi 2013). The urbanization process is characterized by rapid conversion of vegetation, water features, and agricultural fields into impermeable surfaces (Choudhury et al. 2019). One enduring consequence of this urbanization process is the exponential rise in LST (Al Rakib et al. 2020), resulting from the conversion of vegetated areas to impervious surfaces and the transformation of wetlands to agricultural or waste lands. These variations affect diverse environmental phenomena and exert notable impacts on near-surface atmospheric conditions in urban areas. They play crucial roles in determining energy-water balance, surface reflectivity, temperature distribution, moisture evaporation rates, thermal storage, and wind pattern turbulence. Moreover, they possess the capacity to substantially alter above-ground atmospheric conditions, influencing heat transfer rates to soil and modifying energy and water balance processes (Pal and Ziaul 2017). The consequences of LST extend beyond developed areas to encompass surrounding rural regions as well, accelerating the need for new paradigms toward climate-resilient cities and climate change mitigation measures.

The study of complex correlations between rapid urban development and its subsequent climate effects includes crucial factors in the decline of various environmental indices. Human activities, as major drivers of land use transformations, play significant roles in this process by converting vegetated areas and green covers into impermeable surfaces. Previous studies have emphasized that LULC changes can significantly impact the environment, leading to complex shifts in climate patterns affecting both urban and rural regions (Bieling et al. 2013). Considerable attention has been devoted to the transformation of naturally occurring surfaces into impermeable areas, with specific

emphasis on human-induced modifications (F. Mukherjee and Singh 2020). To better understand LULC changes, scholars worldwide have employed various mathematical indices. The Normalized Difference Vegetation Index (NDVI) is one such measure, frequently used to evaluate how crops and vegetation affect both urban and rural areas (Rashid et al. 2023). The role of NDVI in research underscores its capability to monitor land use changes and their associated impacts (Fathian et al. 2015; Jiang and Tian 2010; OS and AA 2016). This index provides numerical assessment of vegetation conditions, enabling scientists to differentiate and quantify the impact of anthropogenic activities on natural systems and, by extension, on climate fluctuations. Numerous studies have successfully employed NDVI to decipher the intricate relationship between urbanization, land use alterations, and their observable effects on broader climate systems (Hussain et al. 2022). While NDVI is a valuable tool, it can be affected by variables such as soil background, atmospheric conditions, and sensor angles (Li et al. 2021). In urban environments, NDVI has been effectively employed to examine the potential influence of urban green areas on local climatic conditions (Roßberg and Schmitt 2023). However, interpretation should consider the specific context and limitations of the index. Furthermore, scholars have employed the Normalized Difference Water Index (NDWI) (Sun et al. 2012) to identify changes in water bodies by leveraging middle infrared (MIR) and near-infrared (NIR) spectral bands (Hussain et al. 2023). Land use changes are also contingent on the transformation of vegetative cover into developed regions. The Normalized Difference Built-up Index (NDBI) is another mathematical index utilized to combine LULC and LST changes (Hussain et al. 2023; Kafy et al. 2021; Sun et al. 2012).

Bangladesh has witnessed major LULC changes in recent years due to rapid development, causing significant alterations to LULC and LST in the country's central region. Dhaka, the capital city and one of the most densely populated urban agglomerations globally, has been particularly affected (Ahmed et al. 2013). The city and its surrounding areas have experienced marked increases in LST, driven by widespread industrialization and business establishment expansion, which have profoundly influenced land cover across every district (Imran et al. 2021). Furthermore, understanding seasonal LST fluctuations and their impact on LULC dynamics is crucial for reducing climate change risks and designing effective remedies (Mumtaz et al. 2023). Although Dhaka district is among the fastest-growing areas in central Bangladesh, it lacks comprehensive assessment of LULC changes and the correlation between LULC and LST. This gap underscores the urgent need to adopt eco-friendly urban planning approaches and effective climate change adaptation strategies. To address this challenge, it is

vital to undertake research employing remote sensing (RS) methods, which are among the most effective and widely used techniques for analyzing land use and environmental changes (Fathian et al. 2015; Gazi et al. 2021; Imran et al. 2022; Ding and Shi 2013; Fu and Weng 2016;). These studies integrate Google Earth Engine (GEE) platform and Geographic Information Systems (GIS) technologies, which are widely acknowledged for their economic viability in understanding the complex processes of LULC and LST (OS and AA 2016; Nurlina et al. 2023).

Despite the growing body of research on surface temperature assessment and land cover changes, several gaps remain in our understanding of these phenomena in rapidly developing cities of the Global South (Gazi et al. 2021; Imran et al. 2022). Previous studies have often focused on singular aspects of urban climate change, such as land cover transformations or temperature variations, without fully integrating these elements or examining their long-term interplay. Additionally, there is a scarcity of comprehensive, multi-decadal analyses that capture dynamic trends in urban expansion and associated thermal consequences in megacities like Dhaka. Our research addresses these gaps by providing a comprehensive, long-term evaluation of land cover changes and their consequences on surface temperature intensity in Dhaka from 1998 to 2022. We employ a novel combination of RS techniques, including the mono-window algorithm (MWA) for LST estimation and advanced land cover classification methods, to analyze a quarter-century of urban transformation. To analyze spatial relationships between LULC changes and surface temperature intensity, we employed both local and global regression techniques, including Ordinary Least Squares (OLS) regression to identify overall patterns and relationships across the entire study area (Zdaniuk 2014). Additionally, we applied Geographically Weighted Regression (GWR) to address spatial non-stationarity and explore local variations in relationships between land cover types and LST (Lessani and Li 2024; Yang 2020). These statistical approaches complement our RS analyses, providing comprehensive interpretation of spatial dynamics in Dhaka's thermal environmental effects. The analytical approach applied in this study allows us to not only quantify changes but also understand complex interrelationships among land cover dynamics, urban expansion, and thermal patterns. The primary goal of this research is to equip decision-makers with comprehensive data crucial for formulating climate change policies and guiding urban and rural development strategies. These findings play crucial roles in informing policies that foster sustainable growth, with emphasis on reducing environmental damage, strengthening social resilience, and promoting the development of smart, healthy cities.

This study offers critical insights for policymakers addressing urban planning challenges, including land-use regulation, ecological preservation, climate adaptation, and zoning strategies. Specifically, the objectives are to: (1) quantify LULC changes in Dhaka from 1998 to 2022 using supervised classification of Landsat imagery; (2) estimate LST variations via the mono-window algorithm; (3) examine the spatial relationships between LULC transitions and LST intensity using both global OLS and local GWR models; and (4) generate actionable, policy-relevant insights to support environmentally sustainable urban development and climate resilience in rapidly urbanizing contexts. The novelty of this research lies in its extended temporal scope, integration of advanced RS techniques, and its focus on a rapidly expanding megacity within a developing country. By incorporating spatial data with multiple climatic variables, the study enables a nuanced analysis of seasonal LST dynamics and their influence on urban land use patterns. Leveraging platforms such as GEE and GIS, the methodology enhances the efficiency, scalability, and interpretability of urban climate assessments. This work contributes a replicable framework for monitoring urban thermal environments and land transformation processes, particularly in densely populated regions of South Asia. By analyzing Dhaka's urban evolution over 24 years, the study captures long-term UHI dynamics often overlooked in shorter-term analyses. Beyond its scientific contributions, the research delivers practical tools and evidence-based recommendations for urban planners and decision-makers. It bridges the gap between empirical research and policy implementation, offering a robust foundation for designing more resilient, sustainable, and livable urban environments in the face of accelerating climate change and urban expansion.

2 Materials and methods

2.1 Study area

Dhaka, the capital of Bangladesh, is one of the most rapidly urbanizing and densely populated regions globally (Fig. 1). Its transformation into a major South Asian megacity has been driven by accelerated economic development and extensive urban expansion (Hu et al. 2021). With one of the highest population concentrations worldwide, Dhaka attracts millions of new residents annually, contributing to significant employment growth and infrastructural pressure. The city's unregulated urbanization has led to widespread land-use changes and environmental degradation.

Geographically, Dhaka is located between latitudes 23.53°N and 28.06°N and longitudes 90.01°E and 90.37°E, encompassing approximately 1,465.60 square kilometers.

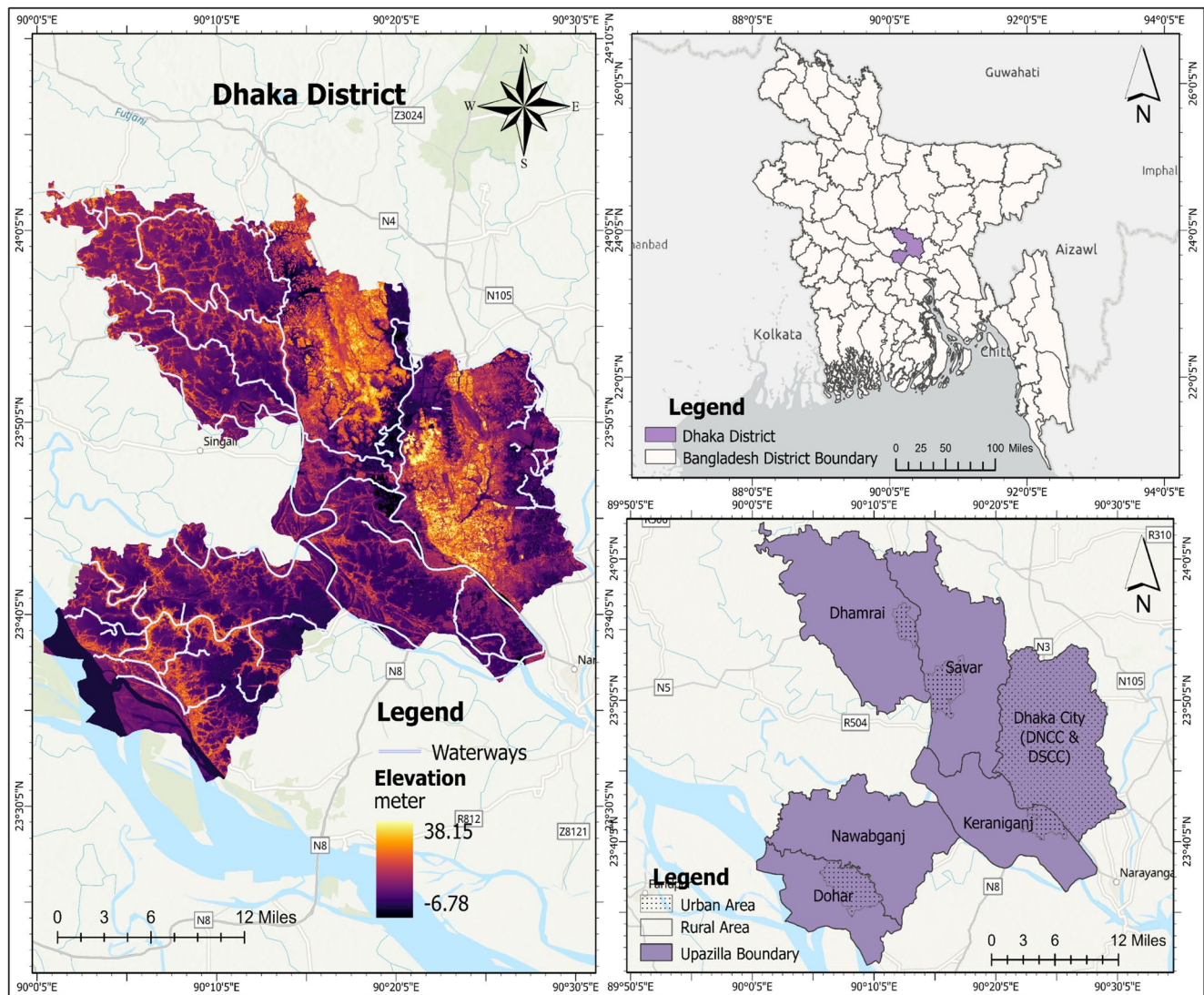


Fig. 1 Geographical location of the study area: Dhaka district, Bangladesh

Its elevation ranges from 2 to 13 m above sea level. As of the latest census, the population stands at 18,305,671, with a density of 12,490 inhabitants per square kilometer (Kafy et al. 2021).

The unchecked expansion of built-up areas has raised concerns among urban and rural policymakers, particularly regarding its environmental consequences. Between 2001 and 2021, built-up land in Dhaka increased from 39 to 66%, while vegetation cover declined sharply from 35 to 6% (Rahman 2022). These rapid LULC transformations have had a pronounced impact on LST dynamics. Understanding the seasonal variability of LST and its interaction with LULC requires a comprehensive, spatially integrated approach. While Dhaka's urban core has undergone significant development, its surrounding rural areas are also experiencing rapid change, contributing to broader LULC shifts and thermal variations. Accordingly, this study incorporates

both urban and rural zones to capture the full spectrum of LST and LULC interactions, offering a holistic perspective on the environmental implications of urbanization in South Asia.

2.2 Datasets

Given the inherent variability in land use patterns and demographic trajectories across urban and rural zones, this study employs a methodological framework capable of capturing seasonal fluctuations in LST and their influence on land use dynamics within the Dhaka district. Recognizing that different regions exhibit distinct seasonal LST trends, the analytical approach is designed to accommodate spatial heterogeneity and temporal variability. To achieve this, a quantitative research design was adopted, enabling robust measurement and interpretation of LST and LULC

Table 1 Description of LULC classification categories used in this study

LULC	Description
Vegetation	Natural and afforested dense forest, grass, or vegetated area
Barren land	Open land, field, playground, newly accreted land
Built-up	Business, factories, residential, transportation, streets, mixed downtown and other developed area.
Waterbody	River, canal, drainage channels, other active meteorological elements

changes. LST data were extracted from Landsat 5 and Landsat 7 satellite imagery for the years 1998, 2004, 2010, 2016, and 2022. All selected images-maintained cloud coverage below 1% and a spatial resolution of 30 m, ensuring high-quality thermal and spectral data for analysis. The study integrates multiple geospatial tools, including GEE and ArcGIS Pro. GEE was utilized for training data selection and validation of LULC classifications, while ArcGIS Pro facilitated demographic data integration, spatial analysis, image classification, and map generation. This multi-platform approach enhances analytical precision and operational efficiency, allowing for comprehensive spatial-temporal assessments of urban and rural transformation. A detailed overview of the datasets and analytical procedures is provided in Tables 1 and 2.

2.3 LULC classification

Our analysis incorporates both LULC aspects, collectively addressed to as LULC. Land cover pertains to the environmental elements covering the surface of the Earth (e.g., water, vegetation, built structures) (Laux et al. 2017), land use denotes the human use of land (e.g., residential, commercial, agricultural) (Arowolo et al. 2018). In urban contexts, land cover often implies land use, but this relationship can be complex. Our classification scheme (Table 1) reflects this integration, with categories like 'Built-up' encompassing both the physical cover (impervious surfaces) and implied use (urban development). We employ established

remote sensing techniques to capture these LULC characteristics, acknowledging the interplay between physical land cover and human land use patterns in shaping urban and rural landscapes.

The Support Vector Machine (SVM) classifier was utilized for land cover classification due to its effectiveness in handling complex and nonlinear class boundaries in remote sensing data (Cengiz et al. 2023; Dziuda 2024; Maulik and Chakraborty 2017). We used the radial basis function (RBF) kernel, which has shown superior performance in land cover classification tasks (Adugna et al. 2022; Awad and Khanna 2015). The SVM variables, which include the negative variable (C) and the kernel width (γ), were calibrated through the grid search process with 5-fold verification to ensure the best classification effectiveness. For training and validation, we collected a total of 500 pixels per class for each year, in total 2000 points, utilizing a combination of high-resolution imagery and field survey data where available. From the collected sample data, 75% was used as training data and the rest of 25% was used as testing points. The land cover classes included in our classification were: built-up area, vegetation, water body, and bare soil (Table 3). These classifications were chosen to represent the major land cover types in the Dhaka metropolitan area and to capture the most significant changes over the study period. Misclassifications were most common between spectrally similar classes, such as bare soil and some built-up areas, particularly in areas of new construction or urban expansion. To address these limitations and improve classification accuracy, we performed post-classification refinements including the application of a 3*3 maximum isolated pixels, and manual corrections of obvious misclassifications based on visual interpretation and local knowledge (Kotzbek 2017). Despite these challenges, our classification results provide a reliable basis for analyzing land cover alteration over the study period. The consistent methodology applied across all years ensures comparability of results, allowing for robust analysis of land cover trends. However, we advise caution in interpreting fine-scale changes, particularly in complex urban areas where classification uncertainties are higher.

Table 2 Descriptions of Landsat satellite imagery used for LULC and LST analysis (1998–2022)

Year	Season	Season	Path/Row	Sensor	Spatial resolution	Cloud cover
1998	Summer	03 April 1998	138/043	Landsat 5 TM	30 m	< 10%
	Winter	26 November 1998				
2004	Summer	19 April 2004				
	Winter	29 November 2004				
2010	Summer	20 April 2010				
	Winter	30 November 2010				
2016	Summer	20 April 2016		Landsat 8 OLI		
	Winter	30 November 2016				
2022	Summer	21 April 2022				
	Winter	23 November 2022				

Table 3 Specifications of Landsat datasets used for LULC classification and LST Estimation (1998–2022)

Year	Dataset	Bands used for LULC and LST	Sensor	Spatial Resolution
1998	LT05_L2SP_137043_19980403_20200909_02_T1	SR Bands 1,2,3,5,7	Landsat 5 TM	30
	LT05_L2SP_137043_19971126_20200909_02_T1			
2004	LT05_L2SP_137043_20040419_20200903_02_T1			
	LT05_L2SP_137043_20041129_20200902_02_T1			
2010	LT05_L2SP_137043_20100420_20200824_02_T1	SR Bands 1,2,3,4,5,6,7	Landsat 8 OLI/TIRS	
	LT05_L2SP_137043_20101130_20200823_02_T1			
2016	LC08_L2SP_137043_20160420_20200907_02_T1			
	LC08_L2SP_137043_20161130_20200905_02_T1			
2022	LC08_L2SP_137043_20220421_20220428_02_T1			
	LC08_L2SP_137043_20221123_20230320_02_T1			

2.4 Accuracy assessment

According to previous studies, a LULC cannot be considered finished until an accuracy evaluation has also been performed (Kullo et al. 2021; Lidzhegu 2012). This is due to the fact that an accurate categorization of satellite images is required to get exact findings from change detection. Consequently, testing the findings is required to determine whether the categorization that was conducted was precise. The precision of such information should be evaluated by comparing the categorized image with reference data, as mistakes occur in most classifications (Alfred Awotwi 2009; Kullo et al., 2021). Within the context of this study, the GEE was used to perform an accuracy assessment. The categorized image data were tabulated to produce a classification error matrix, which was then used to define the level of accuracy achieved (Gunathilaka and Fernando 2022).

$$\text{Total precision (OA)} = \frac{\text{The entire quantity of correctly identified pixels}}{\text{The overall quantity of reference pixels}} * 100 \quad (1)$$

$$\text{User precision (UA)} = \frac{\text{quantity of accurate categorized pixels in each category (diagonal)}}{\text{The overall quantity of reference pixels in each category (total row)}} * 100 \quad (2)$$

$$\text{Producer precision (PA)} = \frac{\text{quantity of accurate categorized pixels in each category (diagonal)}}{\text{The overall quantity of reference pixels in each class (total column)}} * 100 \quad (3)$$

$$\text{Kappa Coefficient (T)} = \frac{\text{The overall number of Sample} * \text{Corrected Sample} - \sum (\text{column.tot} * \text{row.tot})}{(\text{Total number of Sample})^2 - \sum (\text{column.tot} * \text{row.tot})} \quad (4)$$

Table 4, depicts the overall accuracy of urban and rural areas in Dhaka district, the accuracy of the producer, the

accuracy of the user, and Kappa statistics. In 1998, 2004, 2010, 2016, and 2022, the overall accuracy of the categorized images varied between 92.49%, 92.20%, 93.98%, 92.09%, and 90.09%, respectively. The Kappa statistic was discovered to be 0.879, 0.876, 0.912, 0.876, and 0.846 in each of the years from 1998 to 2022. This was in line with the recommended figures (Janssen 1994; Wu et al. 2016) indicated. As a result, more research might be conducted using this data. In addition, a cross-tabulation detection matrix for each category were developed using the change matrix to determine the primary kinds of gains and losses within each category.

2.5 Evaluation metrics

Metrics incorporating preciseness, and F1-score and recall are predominantly dependent upon the correct and incorrect classification of a multitude of instances (Obilor 2018; Qian et al. 2021; Tasnim et al. 2022). Within the scope of research methodology, the numerical values located along the main diagonal bear significance as they represent accurate predictions within a given dataset. Equation (5) explicitly outlines the process of determining accuracy by splitting the total of these precise predictions by aggregating the total number of prognostications made. However, it's imperative to acknowledge that accurately gauging the measure of accuracy is contingent upon effectively handling imbalanced data intricacies. Imbalanced data often introduces challenges such as accuracy confusion, a phenomenon where the traditional accuracy metric might obscure or misrepresent the actual predictive performance

Table 4 Precision evaluation metrics for LULC classification of Dhaka district (1998–2022)

Year	User accuracy (%)				Producer accuracy (%)				Overall accuracy	Kappa statistics
	Built-up	Barren land	Waterbody	Vegetation	Built-up	Barren land	Waterbody	Vegetation		
1998	0.85	0.76	0.97	0.93	0.85	0.76	0.96	0.94	92.49%	0.879
2004	0.92	0.82	0.95	0.89	0.95	0.75	0.96	0.87	92.20%	0.876
2010	0.95	0.90	0.96	0.89	0.96	0.81	0.97	0.91	93.98%	0.912
2016	0.95	0.77	0.93	0.84	0.97	0.68	0.93	0.84	92.09%	0.876
2022	0.94	0.83	0.93	0.74	0.93	0.68	0.91	0.84	90.09%	0.846

due to uneven class distribution. Addressing these complexities becomes crucial to ensure a more nuanced and comprehensive evaluation of predictive models in the face of imbalanced datasets within the research framework. (Qian et al. 2021). This phenomenon arises when the predictive model exhibits a propensity to categorize all instances as members of the dominant class, thereby resulting in a deceptive accuracy metric (Jason Brownlee 2022; Qian et al. 2021). In addition, it is important to inquire about two additional metrics, specifically precision and recall, as illustrated in Eqs. (6) and (7). It is apparent that the occurrence of multiple false positive records can be linked to a decrease in precision, while many false negative data leads to a decline in recall (Qian et al. 2021). The F1-score is a quantitative measure incorporating the considerations of precision and recall. The mathematical representation of this phenomenon can be elegantly expressed as the harmonic mean, as denoted by Eq. (8). All the metrics are limited to the numerical range of zero to one. A numerical value of greater magnitude, tending towards unity, indicates a significant level of effectiveness in the domain of categorization performance (Qian et al. 2021; Tasnim et al. 2022).

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \quad (5)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (6)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

$$\text{F1 score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

Here, TP=True positive, FP=False Positive, TN=True Negative, and FN=False Negative.

In addition, the spatial evaluation domain, within the diverse landscapes encompassing urban and rural area in Dhaka, a comprehensive examination surpasses conventional evaluation indices such as kappa coefficients, UA, and PA. Within the scope of LULC accuracy, the study integrates the elusive yet pivotal Receiver Operating Characteristic (ROC) curve. This enigmatic graph illuminates the interplay between true positive rates (TPR) as denoted by Eq. (9) and false positive rates (FPR) as denoted by Eq. (10) across varying categorization thresholds, depicting a visual narrative rooted in sensitivity and specificity. Leveraging anticipated probabilities of affirmative classifications, the ROC curve serves as a guiding compass, steering towards optimal accuracy amidst this intricate

labyrinth (Cl  men  on and Vayatis 2010). Analogous to celestial cartographers, these curves delineate the intricate trade-offs inherent in a classification model's TPRs and FPRs. The pursuit of superior performance materializes in curves gravitating toward the coveted top-left corner, where the model's prowess reaches its zenith. Central to this plot is the elusive yet influential Area Under the Curve (AUC) as denoted by Eq. (11) a numerical embodiment capturing insights from the multidimensional graph. A higher AUC serves as a testament to superior classification finesse. Embedded within meticulous computations, the ROC curve emerges as a captivating narrative a testament to precision in the ever-evolving landscape of evaluation methodologies (Biswas et al. 2023; Rachakonda and Bhatnagar 2021).

$$\text{TPR} = \frac{TP}{TP + FN} \quad (9)$$

$$\text{FPR} = \frac{FP}{FP + TN} \quad (10)$$

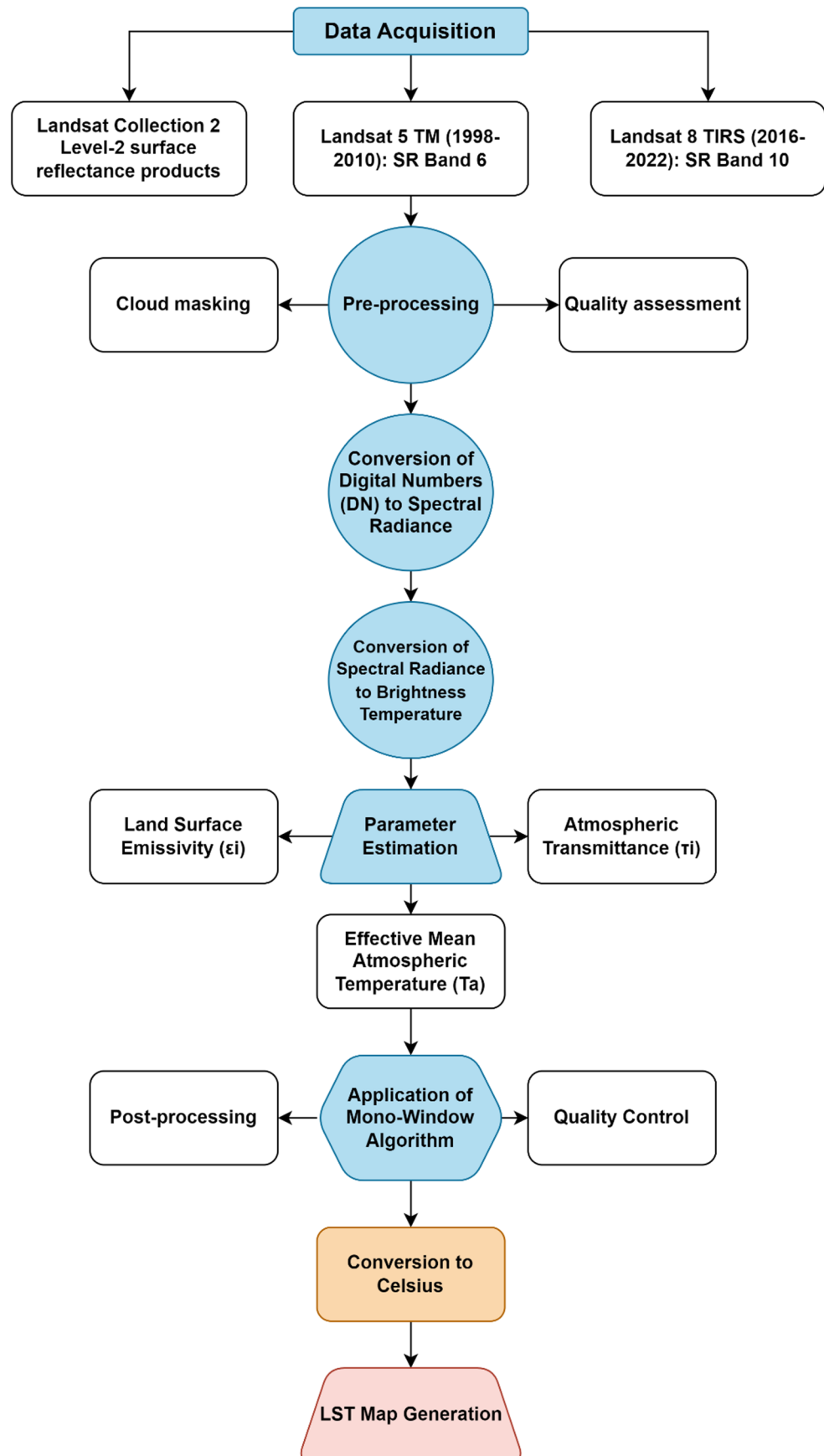
$$\text{AUC} = \int_0^1 \text{TPR} \, d(\text{FPR}) \quad (11)$$

2.6 Estimation of LST

In this research, we utilized surface reflectance products from Landsat Collection 2 Level-2 for LST estimation, utilizing the mono-window approach formulated by (Qin et al. 2001), was chosen for its balance of accuracy and simplicity, as well as its applicability across different Landsat sensors (Jiang and Lin 2021; Sekertekin and Bonafoni 2020). This algorithm has been widely used in thermal characteristics studies and has shown good performance in tropical urban environments (Mukherjee and Krishna 2024; Wang et al. 2015). While ready-to-use LST products are available, implementing the mono-window algorithm ourselves allowed for greater control over processing parameters and better adaptation to our specific study area. This approach enhanced the transparency and reproducibility of our methodology. We performed rigorous quality control, including cloud masking and outlier removal, to ensure the reliability of our LST estimates. Table 2, provides details of the Landsat datasets used, including their spatial and spectral characteristics. Figure 2, illustrates our LST estimation workflow, demonstrating how we integrated the Landsat Collection 2 data with the mono-window algorithm to develop consistent and accurate LST maps across our study period.

For our analysis, we used thermal infrared data from Landsat 5 TM (1998–2010) and Landsat 8 TIRS

Fig. 2 Flowchart illustrating the seasonal LST assessment process using the mono-window algorithm



(2016–2022) sensors. Specifically, we utilized SR Band 6 (10.40–12.50 μm) for Landsat 5 and SR Band 10 (10.60–11.19 μm) for Landsat 8. Three parameters are needed for mono-window algorithm: LSE was estimated using the NDVI thresholds method, considering atmospheric transmittance and effective mean atmospheric temperature. (Sobrino et al., 2008). The transmittance and mean temperature of the atmosphere were calculated based on local meteorological data obtained from nearby weather stations. The process involves several steps (N. R. Mukherjee and Krishna 2024; Qin et al. 2001; Wang et al. 2015):

- i. Transformation of Digital Numbers (DN) to Spectrum Brightness: First, we converted the DN values of the thermal bands to spectral radiance using the Eq. (12):

$$L = L_{\min} + (L_{\max} - L_{\min}) * Q_{\text{dn}}/Q_{\max} \quad (12)$$

Where: L is the spectral radiance $L_{\max}$ and $L_{\min}$ are the lowest and highest thermal brightness values Q_{dn} is the identification number of the pixel $Q_{\max}$ is the maximum possible DN value.

- ii. Calculation of Temperature Conditions from Spectrum Radiance: Next, we converted the spectral radiance to brightness temperature via the inverse application of the Planck function (Eq. 13):

$$T_i = K_2 / \ln(1 + K_1 / L) \quad (13)$$

Where: T_i is the brightness temperature in Kelvin K_1 and K_2 are calibration constants specific to the Landsat sensor L is the spectral radiance.

- iii. Application of the Mono-Window Algorithm: The core equation of the mono-window algorithm is (Eq. 14):

$$T_s = [a_i(1 - C_i - D_i) + (b_i(1 - C_i - D_i) + C_i + D_i)T_i - D_iT_a] / C_i \quad (14)$$

Where: T_s is LST ($^{\circ}\text{C}$); T_i is brightness temperature; T_a is mean atmospheric temperature; T_a , T_a are Planck function constants; C_i , D_i are internal parameters using Eq. (15):

$$C_i = i * i D_i = (1 - i) * [1 + (1 - i) * i] \quad (15)$$

Where: τ_i is the atmospheric transmittance ε_i is the LSE.

- iv. Estimation of Parameters.

- a) Land Surface Emissivity (ε_i): We estimated LSE employing the NDVI threshold technique (Sobrino

et al. 2008). This involves calculating NDVI and assigning emissivity values based on NDVI ranges.

- b) Atmospheric Transmittance (τ_i): We calculated atmospheric transmittance using a linear equation based on water vapor content (Eq. 16):

$$\tau_i = c - d * w \quad (16)$$

Where c and d are coefficients specific to the Landsat sensor, and w is the atmospheric water vapor matter.

- c) Effective Average Air Temperature (T_a): We estimated T_a utilizing Eqs. 17 and 18:
For center of the latitude in summers:

$$T_a = 16.01210 + 0.92621T_o \quad (17)$$

and for mid-latitude winters:

$$T_a = 19.2704 + 0.91118T_o \quad (18)$$

Where T_o is the near-surface temperature in Kelvin. LST in degrees Celsius was achieved using Eq. (19)

$$LST = T_o \text{ summer and winter} - 273.15 \quad (19)$$

- v. Quality Control and Post-processing: After applying the mono-window algorithm, we performed quality control measures, including cloud masking and outlier removal, to ensure the reliability of our LST estimates.

The mono-window algorithm approach was applied consistently across all years of our study period (1998–2022), using Landsat 5 TM Band 6 for earlier years and Landsat 8 TIRS Band 10 for later years, resulting LST maps. While we acknowledge the existence of ready-to-use LST products, implementing the mono-window algorithm ourselves allowed for greater control over processing parameters and better adaptation to our specific study area. Moreover, this approach enhanced the transparency and reproducibility of our methodology. To account for potential differences between Landsat 5 and Landsat 8 data, we performed cross-sensor calibration using overlapping scenes from the two sensors. We also applied rigorous quality control measures, including cloud masking and outlier removal, to ensure the reliability of our LST estimates. By implementing this algorithm ourselves, we maintained control over all processing parameters, allowing for adaptation to our specific study area and ensuring consistency across the multi-decadal analysis.

2.7 Population and Growth Analysis (PGA)

PGA analysis measures the concentration of people within a defined geographical area relative to its overall land-mass, incorporating both urban and rural sectors (Miah et al. 2024). This density is typically expressed in terms of individuals per square kilometer or per acre (Angel et al. 2007). For example, a city having the same population but occupying a smaller area would look denser and less expensive than one covering a larger area. This density metric is instrumental in evaluating the sprawl of urban and rural territories, shedding light on land utilization within a given region as a cohesive unit. Conventional mathematical and statistical methods often struggle to encapsulate the nuances of spatial data, which inherently differ from straightforward statistical information (Ren et al. 2013). This distinction underscores the necessity for integrating traditional analytical approaches with advanced spatial technologies, such as ArcGIS Pro, to navigate the complexities of spatial pattern dynamics effectively (Angel et al. 2007; Batty 2000). Such integrative approaches have evolved to offer more nuanced insights into urban expansion trends within metropolitan areas, assuming adequate geographic data are available (Ren et al. 2013). Utilizing data from the BBS, alongside other indices developed by Jiang et al. (2007) and others, facilitates a comprehensive analysis of urban and rural growth patterns in districts like Dhaka. Such indices are pivotal for processing data for cluster and outlier analysis (Jiang et al. 2007; Sudhira et al. 2004). These indices are pivotal for processing data for cluster and outlier analysis (Punia and Singh 2012), which, when combined together by the capabilities of the ArcGIS Pro environment, enable a detailed examination of land use change both spatially and periodically (Punia and Singh 2012).

2.8 OLS analysis

In our study, we employed the OLS method, a widely utilized multiple-regression analysis technique (Luo and Peng 2016). This technique assumes spatially constant properties of both independent and dependent variables throughout the defined research region, encompassing both urban and rural

locations. OLS, as highlighted by (Soltanifard et al. 2022), exhibits versatility, rendering it suitable for various datasets and applicable uniformly across the entire study area. The OLS model, articulated by (Luo and Peng 2016), is expressed as follows:

$$\gamma_i = \beta_0 + \sum_k \beta_k x_{ik} + \varepsilon_i \quad (20)$$

In this study, γ_i is the dependent variable, x_{ik} is the independent variable, β_0 is the intercept, β_k is the regression coefficient, and ε is the random error. OLS regression enabled direct assessment of relationships between spatial indices, elevation, slope in urban and rural areas, and seasonal LST, facilitating comparison with GWR findings (Miah et al. 2024).

2.9 GWR assessment

To investigate spatial variability in the relationship between LST and land cover characteristics, this study employed the GWR model. GWR enables localized modeling by allowing regression coefficients to vary across geographic space, thereby capturing spatial heterogeneity in the data (Soltanifard et al. 2022). The model was applied to estimate LST patterns and their associations with land use and topographic variables across urban and rural zones of the Dhaka district. The GWR model is expressed as:

$$\gamma_0 = \beta_0(u_i v_i) + \sum_k \beta_k(u_i v_i) x_{ik} + \varepsilon_0 \quad (21)$$

where $(u_i v_i)$ are the spatial coordinates of location i , $\beta_0(u_i, v_i)$ is the location-specific intercept, $\beta_k(u_i, v_i)$ are the spatially varying coefficients for the independent variables (x_i), and (e_i) is the residual error assumed to follow a normal distribution (Luo and Peng 2016).

The analysis was conducted using ArcGIS Pro, incorporating multiple explanatory variables including NDVI, NDBI, NDWI, NDBSI, elevation, and slope (Table 5). These variables were selected to reflect both biophysical and topographic influences on LST from 2016 to 2022. The model performance was evaluated using adjusted Akaike Information Criterion (AICc) and coefficient of determination (R^2), allowing comparison with the global Ordinary Least Squares (OLS) model (Brunsdon et al. 2010). To assess spatial autocorrelation in model residuals, Global Moran's I was calculated. This index ranges from -1 (perfect dispersion) to $+1$ (perfect clustering), with values near zero indicating random spatial patterns. Ensuring that residuals are spatially independent is critical for validating model reliability (Zhao et al. 2018). The GWR model provided detailed insights

Table 5 Equations for spatial indices used in the study

Name of and equations	References
$NDVI = \frac{NIR - RED}{NIR + RED}$	(Abutaleb et al. 2015; Gascon et al. 2016)
$NDWI = \frac{GREEN - NIR}{GREEN + NIR}$	(Özelkan 2020)
$NDBI = \frac{SWIR - NIR}{SWIR + NIR}$	(Arekhi et al. 2019)
$NDBSI = \frac{(RED + SWIR) - (NIR + BLUE)}{(RED + SWIR) + (NIR + BLUE)}$	(Abutaleb et al. 2015; Ngandam Mfondoum et al. 2016)

into localized drivers of seasonal LST variation, revealing spatial patterns of influence across Dhaka's urban and rural landscapes. A visual summary of the methodological workflow and analytical framework is presented in Fig. 3.

3 Results

3.1 Evaluation of LULC model precision

The accuracy assessment of the LULC classification model demonstrated strong and consistent performance across both urban and rural areas of the Dhaka district from 1998 to 2022. As detailed in Table 6, the classification achieved high

precision across all land cover categories, with particularly strong results for built-up areas and water bodies. Precision values ranged from 0.894 to 0.978, recall values from 0.881 to 0.957, and F1-scores from 0.943 to 0.965 throughout the study period. The AUC metrics (Table 6) further validate the robustness of the classification. AUC values consistently exceeded 0.80 for all land cover classes, with many surpassing 0.90, indicating excellent discriminatory power.

Figure 4 illustrates the Receiver Operating Characteristic (ROC) curves, which show a strong alignment with the top-left corner of the plot across all classes and years—further confirming the high accuracy of the classification model. The overall classification accuracy for built-up areas, water bodies, vacant land, and vegetation averaged 96.7%, 92.2%,

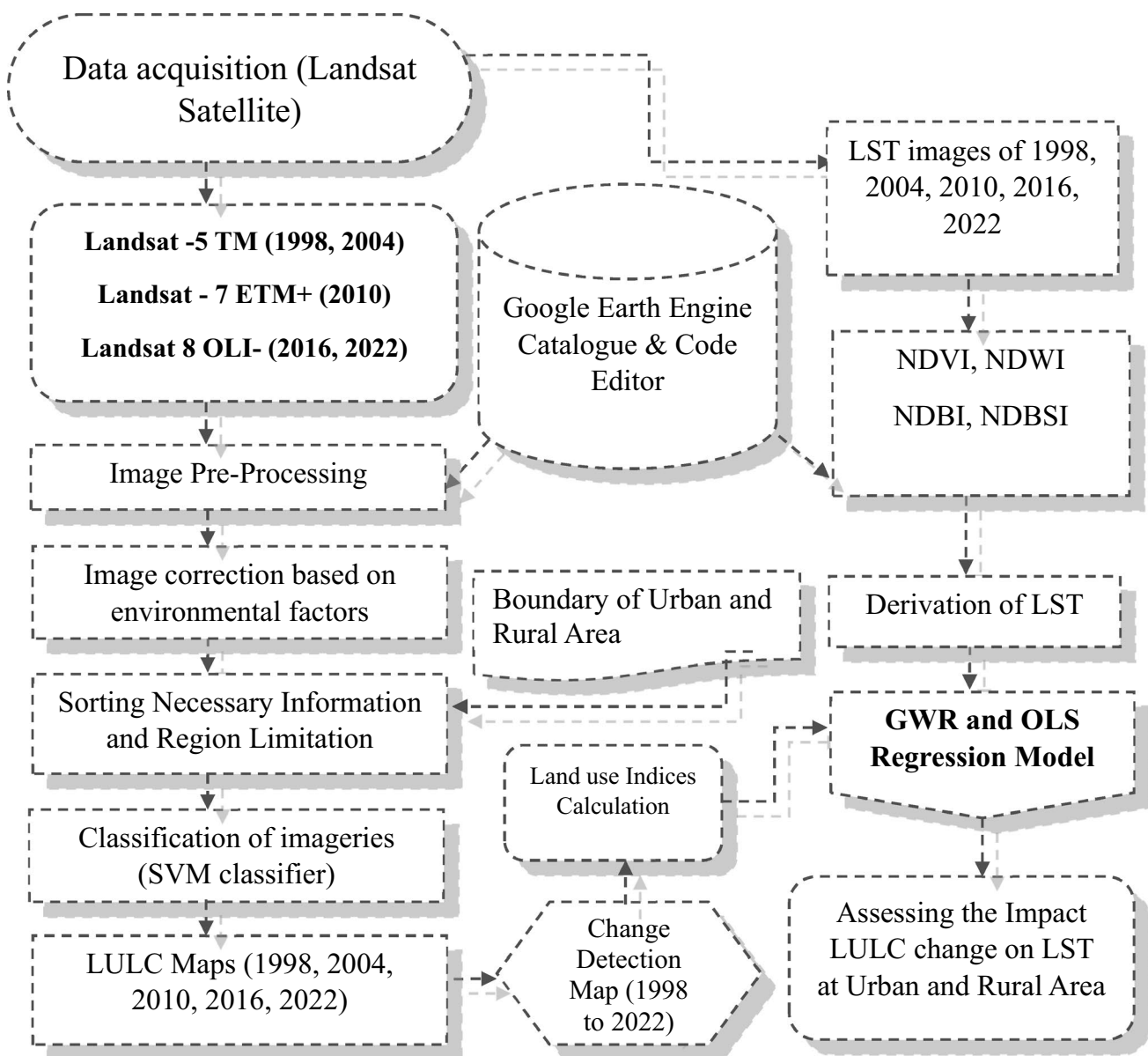
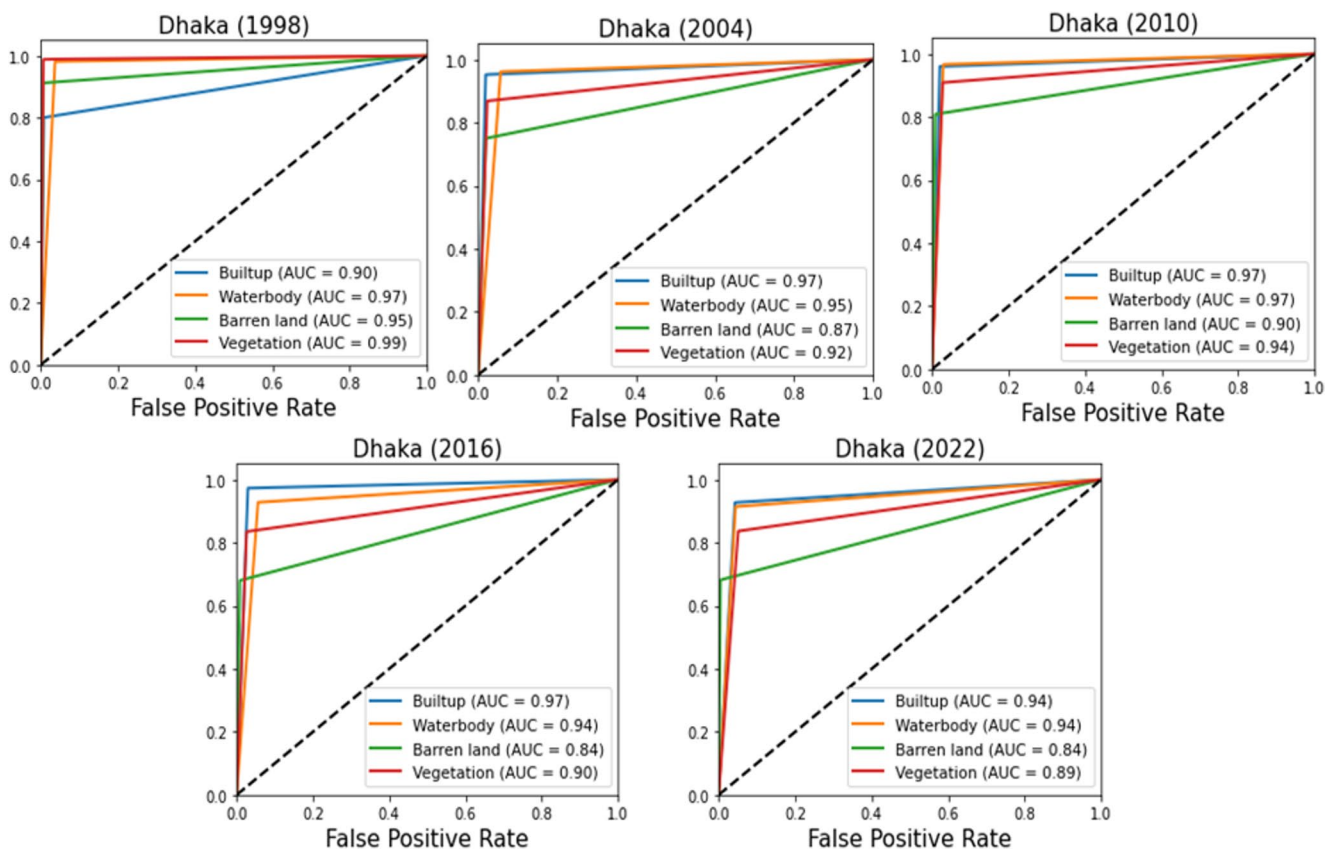


Fig. 3 Methodological flowchart depicting the analysis of seasonal LST impacts on LULC in urban and rural areas of Dhaka district

Table 6 Classifier validation parameters for LULC classification images (1998–2022)

Year	Indicator	Built-up	Waterbody	Barren land	Vegetation	Precision	Macro avg	Overall avg
1998	Efficiency	1.000	1.000	1.000	1.000	0.957	0.921	0.974
	recollect	0.969	0.962	0.969	0.833	0.957	0.887	0.971
	F1- rating	0.950	0.926	0.984	0.909	0.957	0.835	0.971
	AUC	0.880	0.943	0.984	0.916			
2004	Efficiency	1.000	1.000	0.963	0.714	0.942	0.978	0.929
	recollect	0.833	0.860	0.988	0.877	0.942	0.964	0.929
	F1- rating	0.909	0.887	0.975	0.870	0.942	0.971	0.927
	AUC	0.916	0.784	0.979	0.821			
2010	Efficiency	0.962	0.969	0.974	1.000	0.903	0.985	0.969
	recollect	0.926	0.950	0.974	0.938	0.903	0.960	0.969
	F1- rating	0.943	0.880	0.974	0.968	0.903	0.971	0.969
	AUC	0.959	1.000	0.978	0.968			
2016	Efficiency	0.936	1.000	0.950	0.917	0.901	0.936	0.924
	recollect	0.863	0.875	0.894	1.000	0.901	0.924	0.921
	F1- rating	0.898	0.933	0.921	0.957	0.901	0.927	0.920
	AUC	0.922	0.937	0.932	0.987			
2022	Efficiency	0.934	1.000	0.833	0.739	0.931	0.949	0.934
	recollect	0.947	0.850	0.882	0.857	0.931	0.937	0.931
	F1- rating	0.956	0.897	0.850	0.885	0.931	0.945	0.932
	AUC	0.893	0.788	0.838	0.892			

**Fig. 4** ROC curves for LULC classification accuracy assessment in Dhaka district (1998–2022)

94.0%, and 92.1%, respectively. Weighted averages across the study years remained similarly high, reinforcing the model's reliability. These consistently strong performance metrics underscore the effectiveness of the classification algorithm in capturing the complex and evolving land cover patterns of the Dhaka district. This high level of accuracy provides a solid foundation for subsequent analyses of LULC dynamics and their relationship with LST.

3.2 LULC dynamics and Spatial change detection in Dhaka (1998–2022)

Our analysis of LULC changes in Dhaka from 1998 to 2022 reveals a dramatic transformation of the urban and rural landscape. As shown in Fig. 5, urban areas experienced a substantial 98% increase in develop areas, coupled with significant reductions in plant cover (48%) and water bodies (19%). This urban expansion reflects the city's rapid growth and increasing demand for housing and infrastructure. Table 7, provides a detailed breakdown of these changes, highlighting those

rural areas underwent equally significant transformations. Built-up areas in rural zones increased by 120%, indicating rapid rural development. However, this growth was accompanied by a 30% reduction in water bodies and a 62% decrease in barren land. Interestingly, vegetation cover in rural areas showed more resilience, with only a 19% decrease over the time. Although these shifts appear spatially, illustrated in Fig. 6, were not uniform across the district. The areas of Dhaka district that are in its northwest and southeast had the most rapid urbanization, while the southern areas showed a more gradual transition. Rural areas closest to the city boundaries experienced the most significant transformations, suggesting an expanding urban influence.

Figure 7 provides a detailed visualization of the change detection in LULC from 1998 to 2022 in both urban and rural areas of Dhaka district. This map clearly illustrates the geographical patterns of landscape transitions over the study period. In urban areas, we can observe a significant expansion of built-up areas, particularly in the city center and along vital shipping corridors. It also shows how water and other

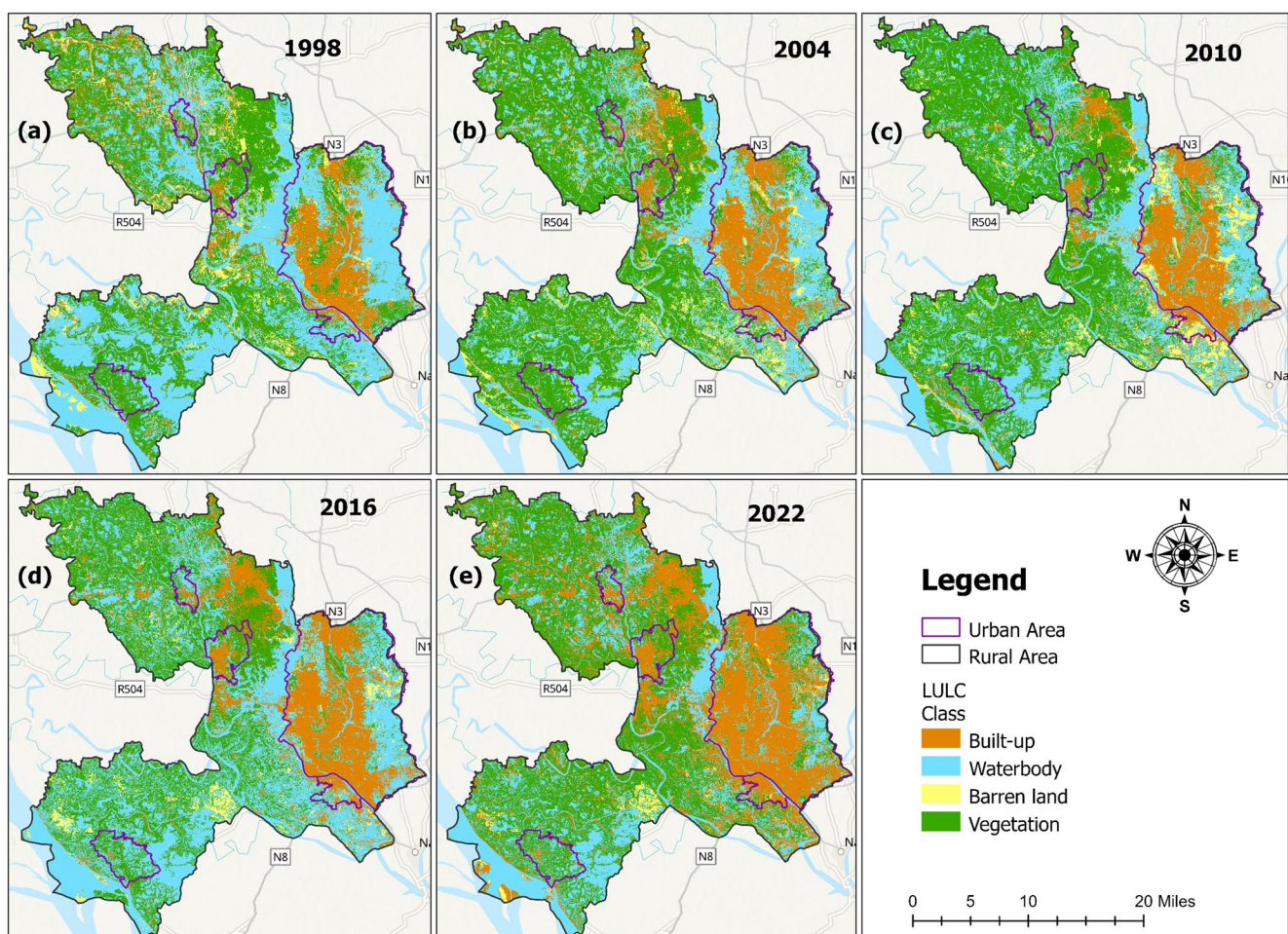
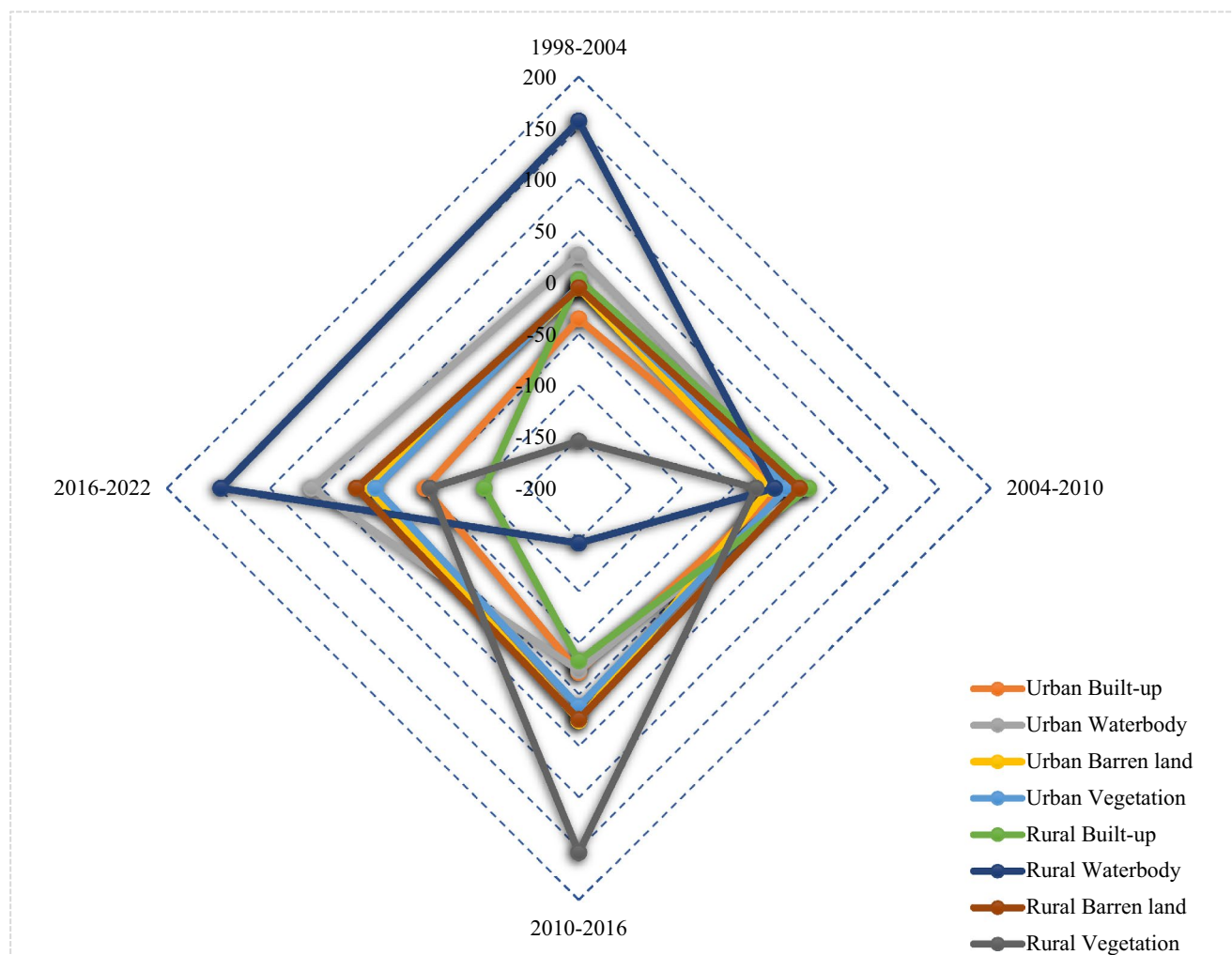


Fig. 5 Spatiotemporal changes in LULC across urban and rural areas of Dhaka district (1998–2022)

Table 7 LULC from 1998–2022 in Urban-Rural area (Km²)

	Year LULC	1998	2004	2010	2016	2022	1998–2004	2004–2010	2010–2016	2016–2022
Urban	Built-up	113.55	149.04	153.19	174.44	225.16	−35.49	−4.15	−21.25	−50.72
	Waterbody	166.19	139.72	121.6	144.45	84.82	26.47	18.12	−22.85	59.63
	Barren land	15.27	20.99	35.46	9.68	5.72	−5.72	−14.47	25.78	3.96
	Vegetation	42.84	46.13	43.86	31.86	34.44	−3.29	2.27	12	−2.58
Rural	Built-up	94.57	91.75	68.43	100.65	208.92	2.82	23.32	−32.22	−108.27
	Waterbody	490.91	334.10	343.86	490.50	342.91	156.81	−9.76	−146.64	147.59
	Barren land	71.04	76.32	62.00	37.29	21.34	−5.28	14.32	24.71	15.95
	Vegetation	413.63	567.99	595.87	441.71	496.99	−154.36	−27.88	154.16	−55.28

**Fig. 6** Net changes in LULC categories (km²) for urban and rural areas of Dhaka district across four time periods (1998–2022)

ecological features are decreasing from the depicted area in these rapidly urbanizing zones. In rural areas, the change is more heterogeneous, with patches of new built-up areas emerging amidst agricultural lands and natural vegetation. The figure effectively captures the dynamic nature of land cover change, showing areas of stability (unchanged colors) and areas of dramatic transformation (color changes). This

spatial representation of land cover change complements the quantitative data presented in Table 7, providing a comprehensive view of the land use dynamics in Dhaka over the past 20 years.

Table 8, presents a detailed breakdown of LULC changes in both urban and rural areas of Dhaka district from 1998 to 2022, quantified in km². This table offers valuable insights

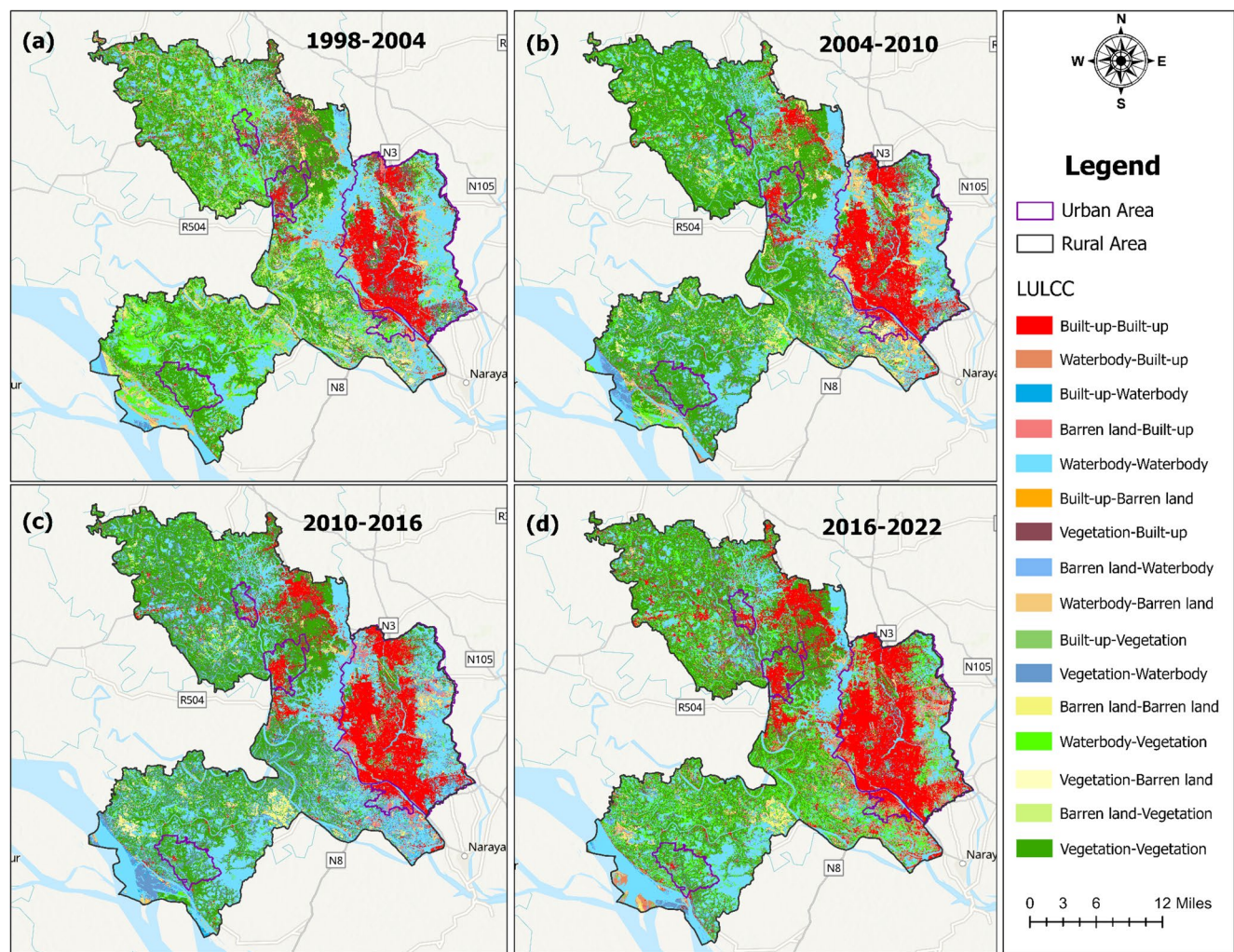


Fig. 7 Spatial distribution of LULC changes in urban and rural areas of Dhaka district (1998–2022)

Table 8 Detailed LULC change matrix (km²) for urban and rural areas of Dhaka district across four time periods (1998–2022)

LULC	Urban				Rural			
	1998–2004	2004–2010	2010–2016	2016–2022	1998–2004	2004–2010	2010–2016	2016–2022
Built-up-Built-up	102.71	128.32	140.86	165.35	22.44	30.73	39.24	74.07
Waterbody-Built-up	11.91	14.27	11.15	37.82	20.23	11.35	10.18	52.35
Barren land-Built-up	4.96	7.88	8.07	4.08	5.85	10.72	16.03	6.96
Vegetation-Built-up	8.49	3.43	12.5	5.43	9.68	5.63	12.26	8.84
Built-up-Waterbody	120.46	94.74	97.94	70.94	287.69	244.71	291.82	276.46
Waterbody-Waterbody	0.69	2.59	0.35	0.63	1.97	5.92	1.53	1.24
Barren land-Waterbody	25.93	7.17	9.92	16.57	39.40	20.73	38.98	73.66
Vegetation-Waterbody	0.67	4.02	13.48	1.71	5.61	17.21	19.73	7.84
Built-up-Barren land	12.69	19.89	4.56	2.98	37.34	16.49	7.55	5.94
Waterbody-Barren land	5.19	10.24	3.91	4.4	64.31	44.37	11.63	18.38
Barren land-Barren land	13.65	14.95	24.97	8.1	34.95	71.21	162.91	51.65
Vegetation-Barren land	3.11	8.21	3.82	1.3	12.08	18.48	11.26	9.86
Built-up-Vegetation	21.13	10.83	7.95	32.72	145.64	61.55	34.31	155.74
Waterbody-Vegetation	4.48	4.78	0.95	0.82	24.92	21.11	16.96	4.29
Barren land-Vegetation	2.99	5.33	5.66	1.24	43.68	35.00	18.75	10.76
Vegetation-Vegetation	49.2	51.61	42.17	34.2	314.36	454.95	377.03	312.11

into the specific transitions between different land cover types over four-time intervals: 1998–2004, 2004–2010, 2010–2016, and 2016–2022. In urban areas, we observe a consistent increase in built-up land across all time periods, with the most significant growth occurring between 2016 and 2022 (50.72 km²). Conversely, waterbodies show a general decreasing trend, with the largest loss (59.63 km²) also occurring in the 2016–2022 period. Barren land and vegetation exhibit more variable patterns, with periods of both increase and decrease. Rural areas show more dynamic changes. Built-up areas increased dramatically by 108.27 km² between 2016 and 2022, indicating rapid rural development in recent years. Waterbodies in rural areas show significant fluctuations, with large decreases in some periods (e.g., 156.81 km² loss from 1998 to 2004) and increases in others. Vegetation cover in rural areas also shows considerable variability, with the largest increase (154.16 km²) occurring between 2010 and 2016. These detailed change statistics provide a nuanced understanding of the land cover dynamics in Dhaka, revealing not just overall trends but also the specific time periods when major transitions occurred. This information is crucial for understanding the pace and features associated with urbanization and rural transformation in the region.

3.3 Demographic shifts and urbanization patterns

Population density and growth trends in Dhaka district show significant variations between urban and rural areas. Figure 8, illustrates that the urban core, particularly Dhaka district, experienced the dramatic population growth. Density increased from approximately 19,000 people/km² in 1998 to over 30,000 by 2022, a growth rate far exceeding the national average.

Urban expansion patterns, depicted in Fig. 9, reveal a clear trend of outward growth from the city center. The areas of Savar, Keraniganj, and Dhamrai showed the highest rates of urbanization outside the DCC area, with population densities increasing by 150–200% over the study period. Rural areas, while experiencing lower overall population growth, showed interesting patterns of development. As evident in Fig. 9, some rural upazilas, particularly those adjacent to urban areas, experienced significant population increases, suggesting a suburbanization trend. For instance, Dohar and Nawabganj saw population growth rates of 50–70% between 1998 and 2022.

3.4 Thermal environment evolution (1998–2022)

Our analysis of LST reveals a clear warming trend across Dhaka district from 1998 to 2022. As illustrated in Fig. 10, urban areas experienced the most significant increases, with

Fig. 8 Changes in population density in urban and rural areas of Dhaka district (1991–2021)

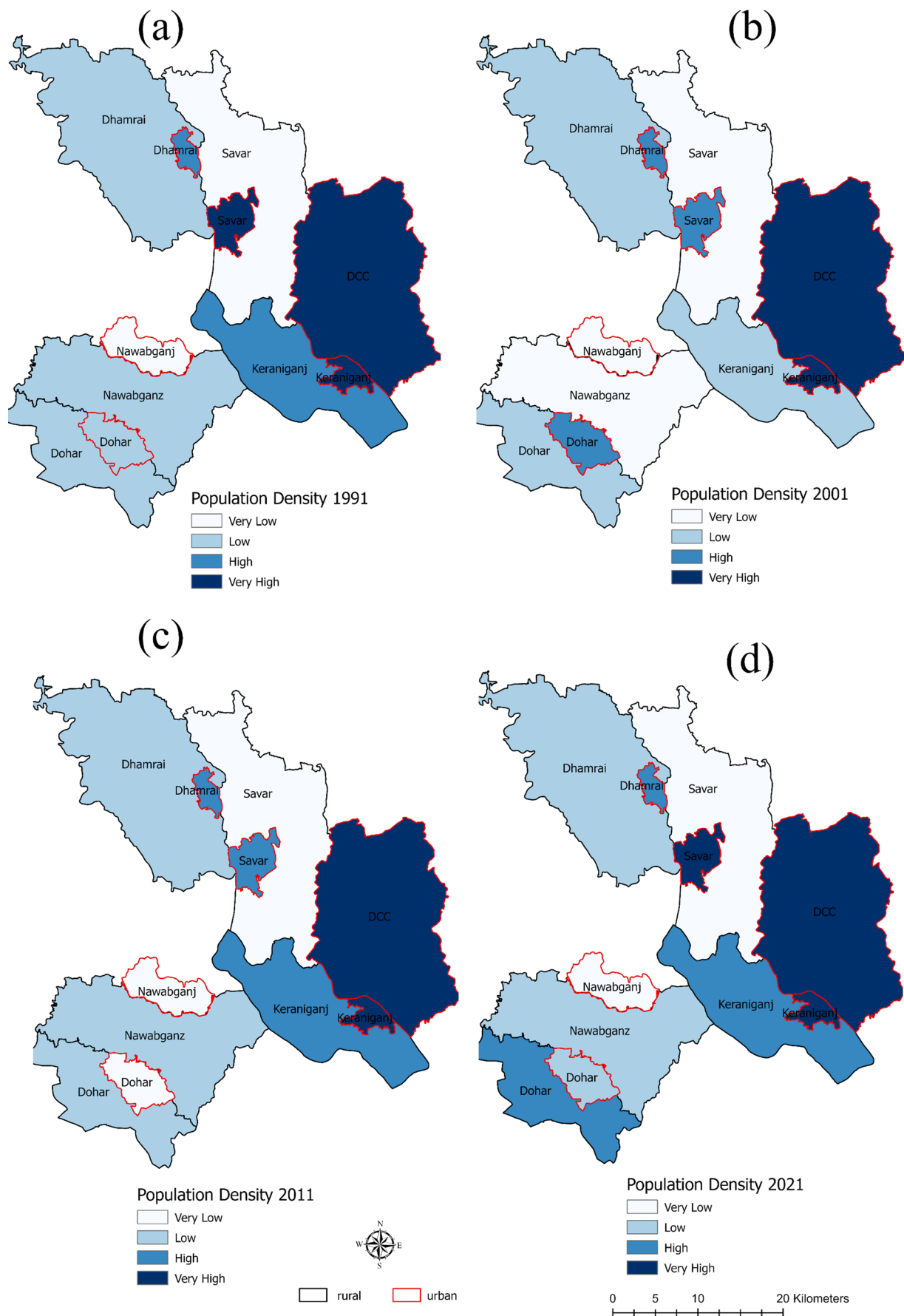
mean LST rising by approximately 3.5°C over the 24-year period. The urban core of Dhaka city consistently showed the highest temperatures, with peak values reaching 36.8 °C in summer months by 2022, as detailed in Table 9.

Figure 11, showcases the seasonal variations in LST, with summer temperatures showing the most dramatic increases. The urban-rural temperature difference, a measure of the UHI effect, was most pronounced during summer nights, reaching up to 6 °C in some areas by 2022. Figure 12, identifies several persistent UHIs, primarily in densely built-up areas with little vegetation cover. The most intense UHI were observed in the old part of Dhaka city and in rapidly developing industrial zones. Conversely, cool spots were associated with large water bodies and remnant green spaces, highlighting their importance in urban temperature regulation.

Accurate estimation of LST is highly dependent on the clear atmospheric conditions and cloud-free research areas. Even with the minimal cloud presence, as low as 10% cloud cover, can significantly impair the accuracy of LST when compared to actual field data (Al Kafy et al. 2020; Chen et al. 2006; Dar et al. 2019; Neteler 2010). Such discrepancies have the potential to introduce distortions in the spatial distribution of LST across different geographic regions, leading to erroneous interpretations. To rigorously validate the predicted LST, we performed a critical comparative analysis using temperature data from the Bangladesh Meteorological Department (BMD) for 1998, 2004, 2010, 2016, and 2022. As shown in Table 9, the analysis revealed certain exceptions in the form of disparities between the LST and actual recorded temperatures. These disparities are categorized as negative disparity (ND) and positive disparity (PD), where ND indicates LST exceeding actual temperatures, while PD indicates LST below actual temperatures. For instance, in 1998, the summer LST exhibited the lowest ND (−0.22), while the winter season of the same year also recorded ND (−0.38), highlighting the consistencies in LST estimation. Conversely, in the summer season of 2016, it has been recorded the highest PD (+2.22) indicating a notable estimation of LST which is strongly matched with the most recent data from 2022, where the mean PD have been observed (+1.35), further emphasizing the consistency in LST measurements with actual recorded values across varying atmospheric conditions.

3.5 Relationship between LULC and thermal characteristics

The effect of develop areas on LST, analyzed through the NDBI, showed a strong positive correlation. As evident in



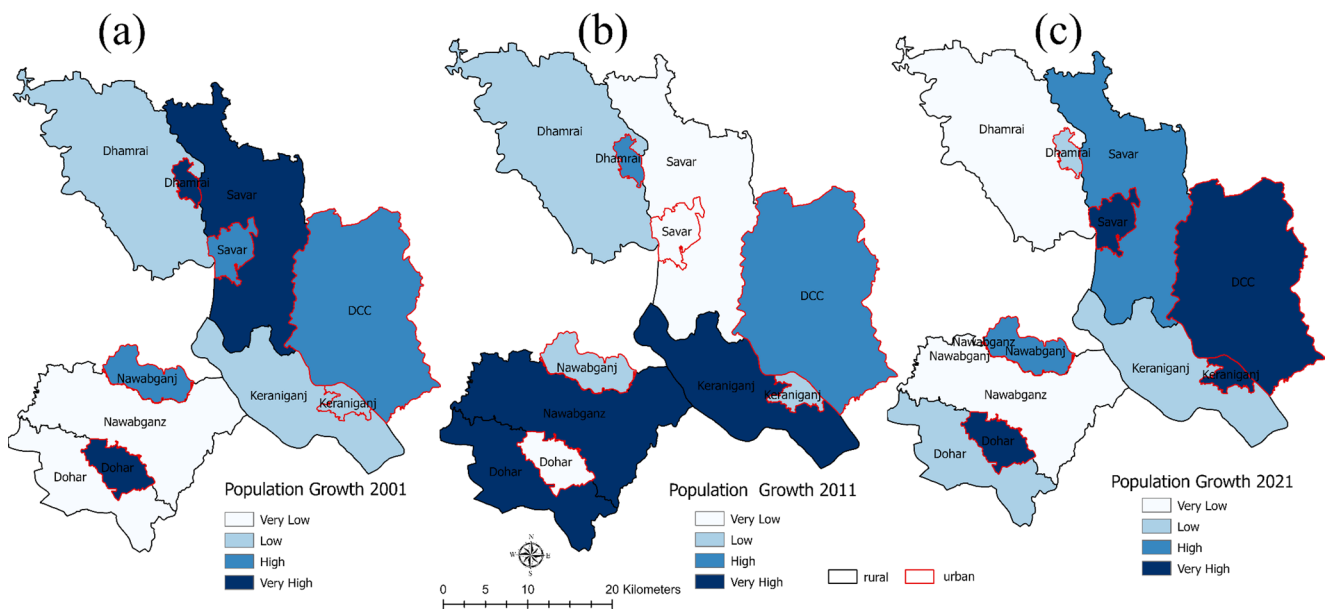


Fig. 9 Population growth trends in urban and rural areas of Dhaka district (1991–2021)

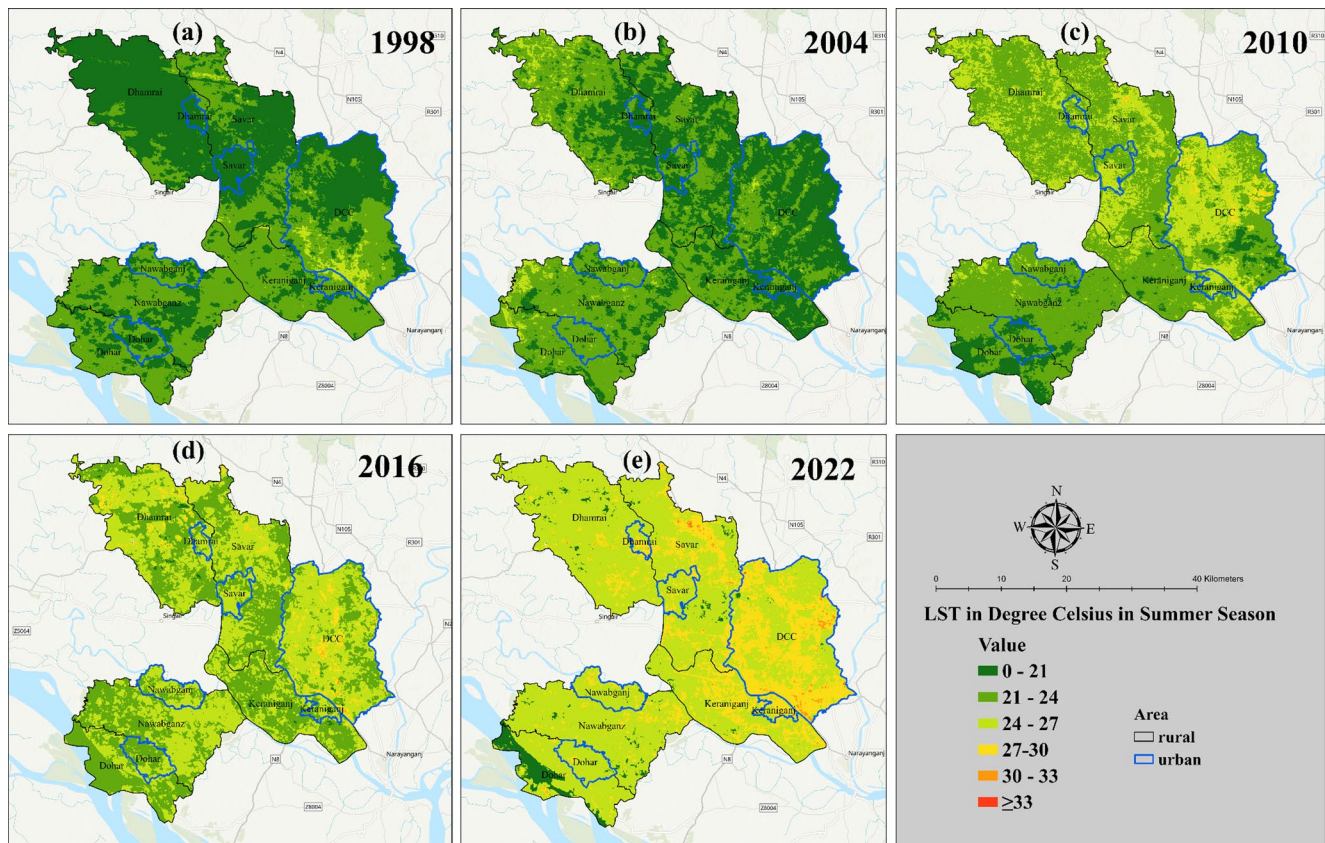


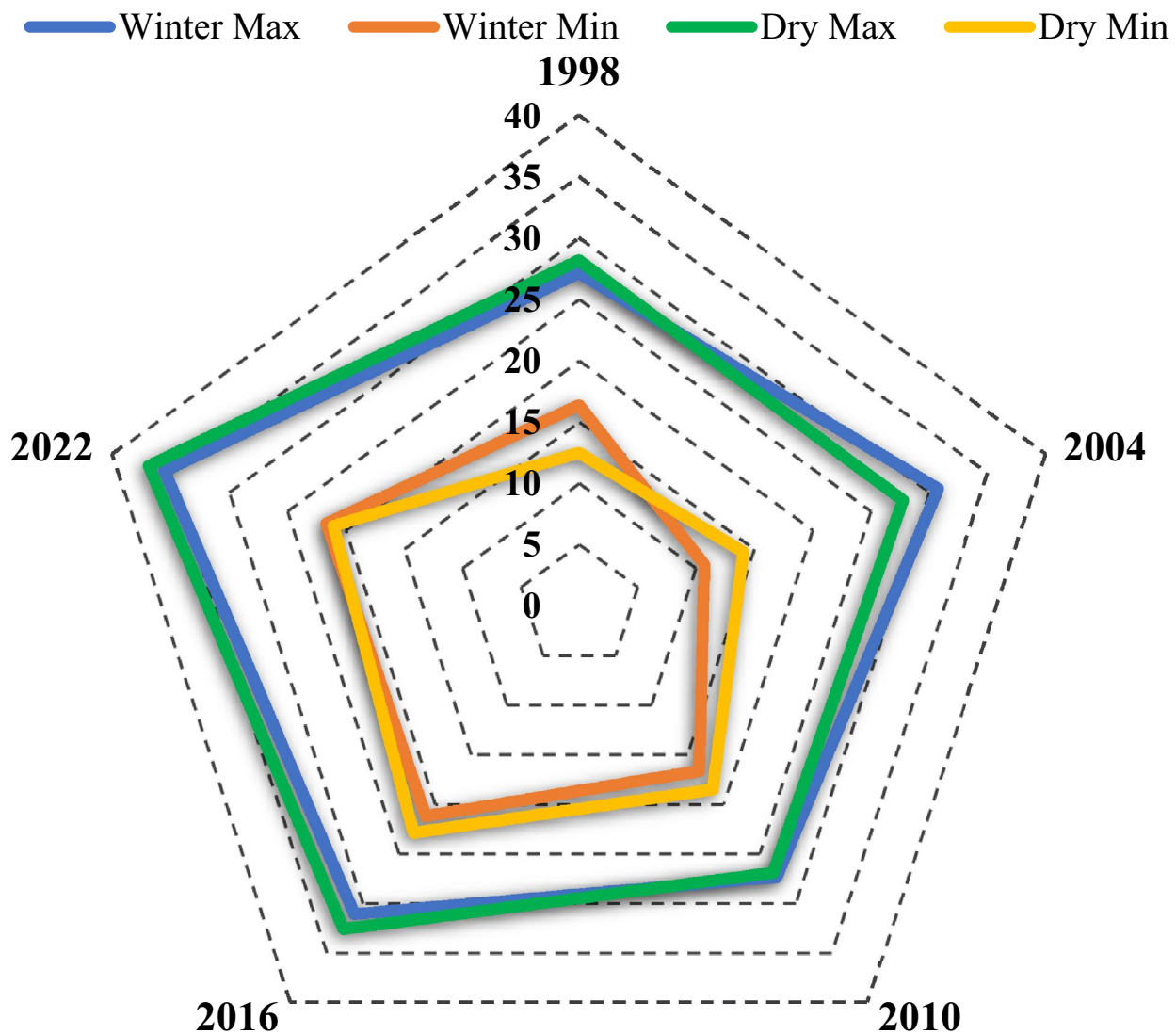
Fig. 10 Spatial distribution of LST in Dhaka district during summer seasons (1998–2022)

Fig. 13, areas with high NDBI values consistently exhibited higher LST, with the relationship strengthening over time. By 2022, a 0.1 raise in NDBI was associated with an average LST increase of 1.5 °C in urban areas. NDBSI showed

a positive correlation with LST, particularly in urban and peri-urban areas. Figure 14, illustrates the changes in NDBSI from 1998 to 2022 across both downtown and countryside areas of the Dhaka district. The figure reveals

Table 9 Validation of seasonal thermal band-derived LST against weather station data for Dhaka district (1998–2022)

Year	1998		2004		2010		2016		2022	
Summer	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min
LST derived from thermal bands(°C)	33.3	16.46	34.8	17.1	34.9	18.9	35.9	22.9	36.8	21.2
BMD observed LST (°C)	34.6	15.6	35.9	16.9	32.5	17.7	34.2	20.2	37.9	22.5
Deviation (°C)	-1.3	+0.86	-1.1	+0.2	+2.4	+1.2	+1.7	+2.7	-1.1	-1.3
Average Deviation	-0.22		-0.45		+1.8		+2.2		-1.2	
Winter	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min
LST derived from thermal bands(°C)	27.13	16.3	30.7	10.7	27.3	16.6	31.1	21.2	35.5	21.6
BMD observed LST (°C)	28.6	15.6	29.9	11.9	29.5	16.2	30.6	20.9	34.9	19.5
Deviation (°C)	-1.47	+0.7	+0.8	-1.2	-2.2	+0.4	+0.5	+0.3	+0.6	+2.1
Average Deviation	-0.38		-0.2		-0.9		+0.4		+1.35	

**Fig. 11** Seasonal variations in maximum and minimum LST (°C) in Dhaka district (1998–2022)

a general increase in NDBSI values over time, especially in urban areas, indicating an expansion of bare or sparsely vegetated surfaces. In 1998, NDBSI values ranged from -0.207 to 0.634 during the winter season and from -0.161

to 0.644 in the summer. By 2022, these ranges had shifted to -0.471 to 0.261 in winter and -0.575 to 0.487 in summer. The higher NDBSI values, particularly evident in the urban core and expanding outwards, correspond to areas of high

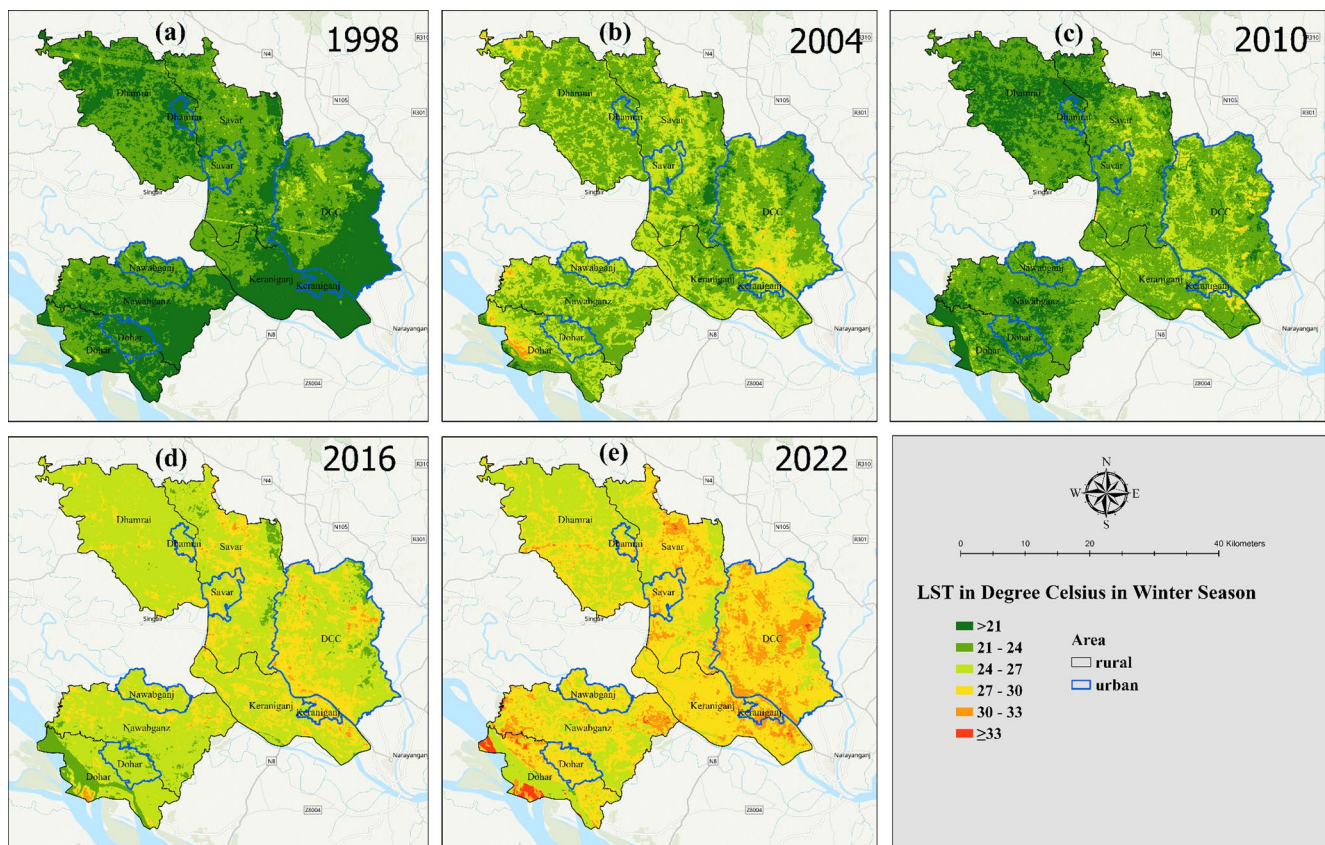


Fig. 12 Spatial distribution of LST in Dhaka district during winter seasons (1998–2022)

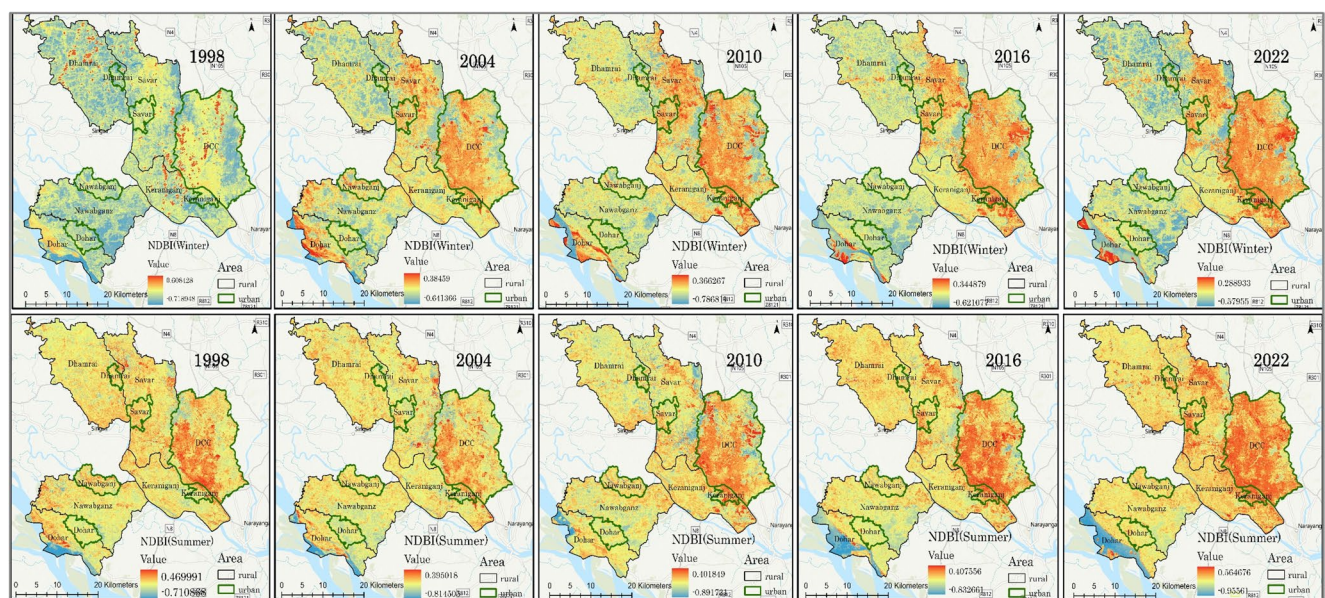


Fig. 13 Temporal changes in NDBI values for urban and rural areas of Dhaka district (1998–2022)

LST. This relationship underscores the heat-absorbing properties of bare soil and impervious surfaces common in urban environments. The temporal increase in NDBSI, especially

in rapidly developing areas on the urban fringe, aligns with the observed rise in LST over the study period. Notably, the summer NDBSI values show more extreme fluctuations

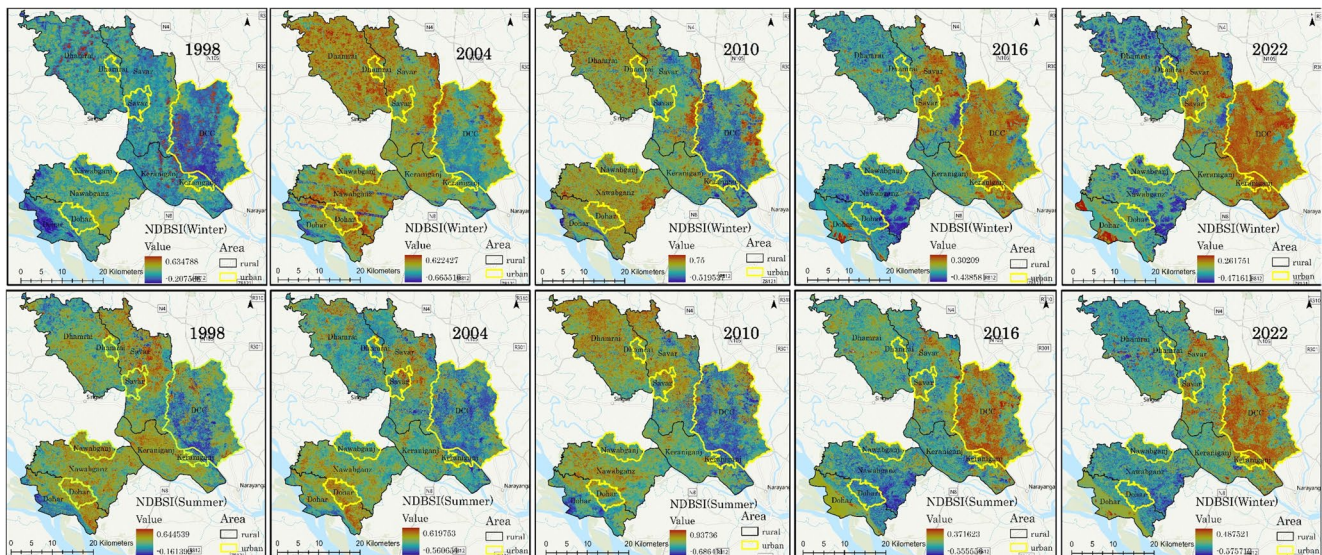


Fig. 14 Temporal changes in NDBSI values for urban and rural areas of Dhaka district (1998–2022)

compared to winter, suggesting a seasonal component to the bareness-LST relationship that may be influenced by factors such as vegetation senescence and soil moisture variations.

Table 10, presents the observed seasonal variations in NDBI and NDBSI across the study period, offering critical insights into how these indices fluctuate between winter and summer from 1998 to 2022. NDBI values exhibit a consistent upward trend, particularly during the summer months. In 1998, the maximum winter NDBI was 0.608, which declined to 0.288 by 2022. Conversely, the summer maximum increased from 0.469 in 1998 to 0.564 in 2022, indicating intensified urban development and expansion of built-up areas during warmer seasons.

NDBSI patterns were more variable. The winter maximum decreased from 0.634 in 1998 to 0.261 in 2022, suggesting a reduction in exposed bare soil or sparsely vegetated surfaces during cooler months. Summer NDBSI values showed greater fluctuation, peaking at 0.937 in 2010 before declining to 0.487 in 2022. These variations may reflect seasonal vegetation cycles and shifts in impervious surface exposure due to urban growth.

Minimum values for both indices trended increasingly negative over time, particularly for NDBI, implying a rise

in areas with minimal built-up or bare soil characteristics—potentially due to the emergence of water bodies or dense vegetation patches. These seasonal dynamics in NDBI and NDBSI underscore the evolving relationship between land cover and LST in Dhaka. The consistent summer increases in both indices align with the observed rise in LST and urban sprawl, reinforcing the link between anthropogenic land transformation and thermal intensification.

Vegetation cover, quantified using the NDVI, demonstrated a significant cooling effect. Figure 15, illustrates those areas with high NDVI values showed lower LST, with the relationship being particularly strong in urban environments. An increase of 0.1 in NDVI was associated with an average LST decrease of 0.8 °C in city areas and 0.5 °C in countryside areas. Water bodies, analyzed using the NDWI, showed a complex relationship with LST, as depicted in Fig. 16. While larger source of water had a cooling effect, smaller water bodies and those in densely built-up areas often showed higher LST, possibly due to pollution and reduced water flow. The cooling effect of water bodies was most pronounced during summer months, with areas within 500 m of large water bodies showing LST up to 3 °C lower than geographical location. Table 11, summarizes the observed

Table 10 Observed seasonal variations in NDBI and NDBSI for Dhaka district (1998–2022)

Year	NDBI				NDBSI			
	Winter		Summer		Winter		Summer	
	Max	Min	Max	Min	Max	Min	Max	Min
1998	0.608	−0.718	0.469	−0.710	0.634	−0.207	0.644	−0.161
2004	0.384	−0.641	0.395	−0.814	0.622	−0.665	0.619	−0.560
2010	0.386	−0.786	0.401	−0.891	0.75	−0.519	0.937	−0.687
2016	0.344	−0.621	0.407	−0.832	0.302	−0.438	0.371	−0.555
2022	0.288	−0.579	0.564	−0.955	0.261	−0.471	0.487	−0.575

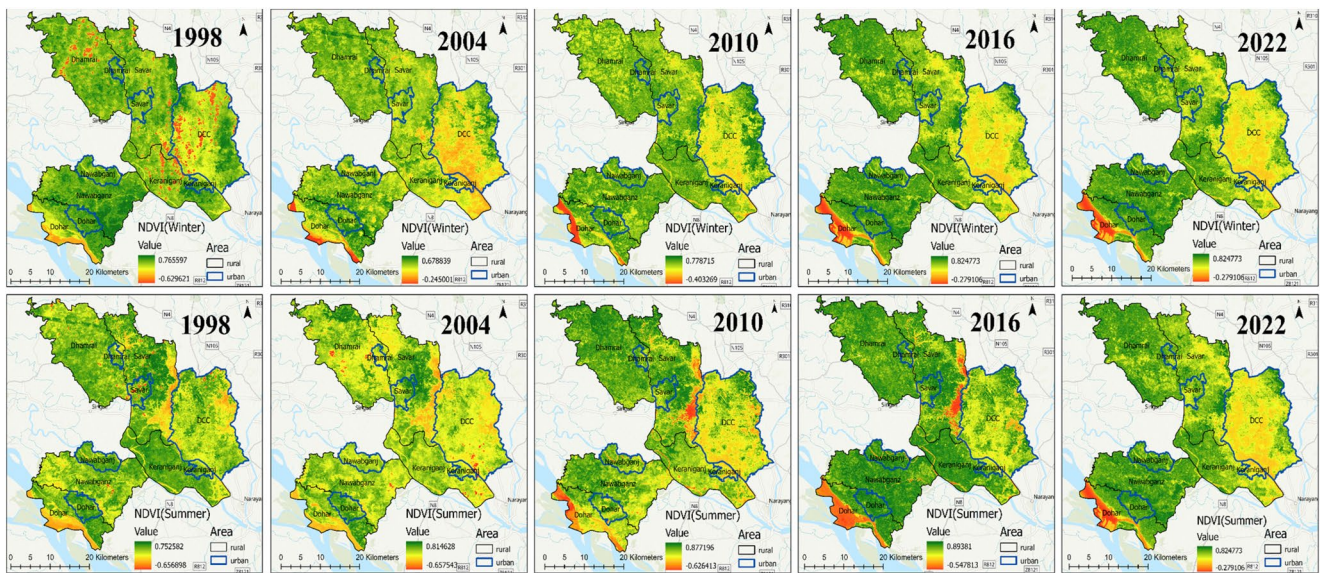


Fig. 15 Temporal changes in NDVI values for urban and rural areas of Dhaka district (1998–2022)

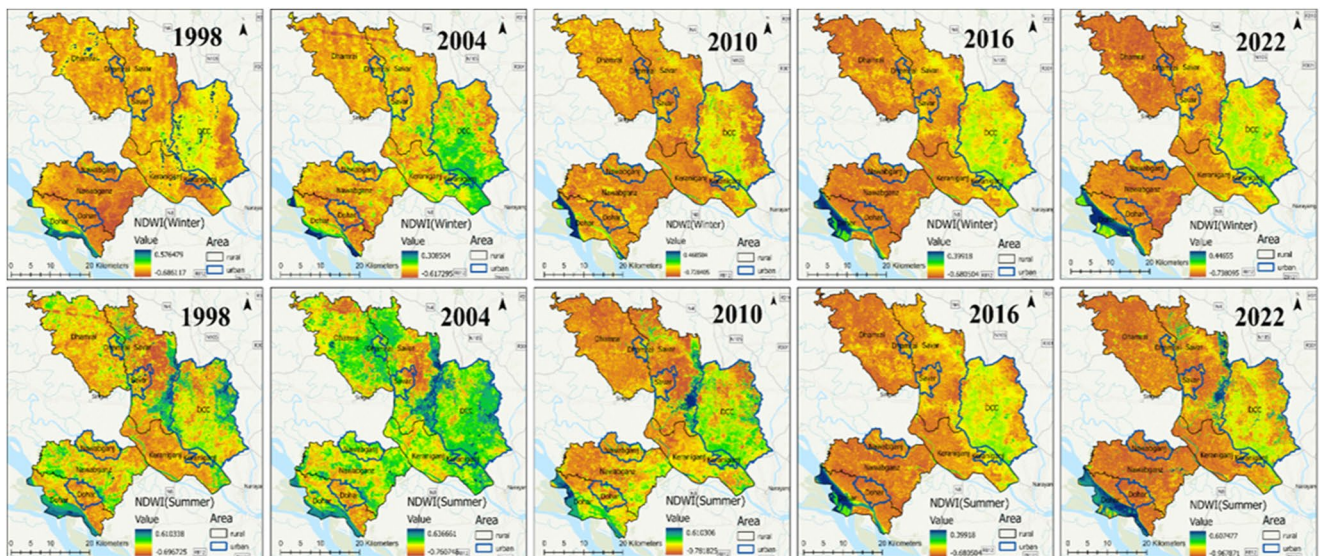


Fig. 16 Temporal changes in NDWI values for urban and rural areas of Dhaka district (1998–2022)

Table 11 Observed seasonal variations in NDVI and NDWI for Dhaka district (1998–2022)

Year	NDVI				NDWI			
	Winter		Summer		Winter		Summer	
	Max	Min	Max	Min	Max	Min	Max	Min
1998	0.752	−0.656	0.752	−0.656	0.576	−0.686	0.610	−0.696
2004	0.678	−0.245	0.814	−0.657	0.308	−0.617	0.636	−0.760
2010	0.778	−0.403	0.877	−0.626	0.468	−0.728	0.610	−0.781
2016	0.824	−0.279	0.893	−0.547	0.393	−0.680	0.399	−0.680
2022	0.832	−0.301	0.912	−0.557	0.446	−0.738	0.607	−0.966

seasonal variations in NDVI and NDWI, providing insight into the dynamic nature of these relationships across different seasons and years. For NDVI, we observe a general

trend of increasing maximum values over the research area, particularly in the summer season. The summer maximum NDVI rose from 0.752 in 1998 to 0.912 in 2022, indicating

an overall increase in vegetation health or density during peak growing seasons. Winter maximum NDVI values also showed an upward trend, from 0.752 in 1998 to 0.832 in 2022, suggesting improved vegetation retention even during cooler months. This trend could be attributed to urban greening initiatives or changes in agricultural practices in rural areas.

Minimum NDVI values fluctuated over the years but generally became less negative, particularly in winter. The winter minimum increased from -0.656 in 1998 to -0.301 in 2022, potentially indicating a reduction in completely barren or highly urbanized areas with no vegetation. NDWI values, on the other hand, show more complex patterns. The maximum NDWI values decreased over time in both seasons, with summer values dropping from 0.610 in 1998 to 0.607 in 2022, and winter values declining more sharply from 0.576 to 0.446. This trend suggests a overall reduction in water body extent or water content in the landscape, which aligns with the observed urbanization trends. Minimum NDWI values became more negative over time, particularly in summer, dropping from -0.696 in 1998 to -0.966 in 2022. This could indicate an increase in very dry or highly built-up areas with minimal water content. The seasonal differences in both indices are noteworthy. NDVI values are consistently higher in summer than in winter, reflecting the natural vegetation cycle. NDWI values, however, show less consistent seasonal patterns, suggesting that water body dynamics in the region are influenced by factors beyond just seasonal precipitation, such as urban development and water management practices. These trends in NDVI and NDWI provide valuable insights into the changing landscape of Dhaka. The increasing NDVI values suggest some positive trends in vegetation cover, possibly due to urban greening efforts or changes in land management. The declining NDWI values urgently highlight severe concerns about the deterioration of water resources and the critical challenges in their management amid rapid urbanization.

3.6 Spatial modeling of LST-LULC relationships

The OLS regression analysis provided insights into the global relationships between land cover indices and LST. As shown in Table 12, the model revealed that NDBI and NDBSI were strongly correlated with LST, while NDVI and NDWI were weakly correlated. The OLS model explained about 57.8% of the LST variation across the entire study area ($R^2 = 0.578$).

The GWR model, however, unveiled significant spatial variations in these relationships. Table 13, compares the performance of OLS and GWR models, showing that the GWR model improved the as an entire model fit, with

Table 12 Coefficient estimates from the OLS regression model

Variable	Coefficient	Standard error	t- statistics	probability
Intercept	9.264	0.622	14.874	0.000000*
NDVI	-91.612	21.266	-4.307	0.00003*
NDBI	43.267	11.107	3.895	0.0001*
NDWI	-47.733	20.854	-2.288	0.023*
NDBSI	31.123	17.165	1.813	0.071
Elevation	0.023	0.040	0.586	0.558
Slope	-0.396	0.259	-1.526	0.036*

* An asterisk appended to a figure signifies statistical significance, denoting that the observed result has a p-value less than 0.05

Table 13 Performance evaluation parameters of GWR and OLS regression models

Parameter	GWR	OLS
R^2	0.9005	0.578
Adjusted R^2	0.836	0.562
AICc	246.854	588.509
Moran's I	Moran's Index	-0.015
	z-score	-0.234
	p-value	0.814

the R^2 increasing to 0.9005. The AICc also decreased significantly from 588.509 in the OLS model to 246.854 in the GWR model, indicating a substantially better model performance.

Figure 17, visually represents the spatial variations in the LST-land cover relationships revealed by the GWR model. It shows that the impact of NDBI on LST was strongest in the densely built-up urban core, with local coefficients up to 50% higher than the global OLS coefficient. The cooling effect of vegetation was most pronounced in areas with contiguous plant area, specifically in some sub urban areas where local coefficients were up to twice the global coefficient. These results highlight the complex and spatially variable nature of the connection between land cover and LST in Dhaka. The findings underscore the importance of considering local context in urban thermal studies and suggest that strategies for mitigating urban heat should be tailored to specific local conditions rather than using a uniform approach throughout the entire urban area.

4 Discussion

4.1 Land cover dynamics and urbanization patterns

The rapid urbanization of Dhaka from 1998 to 2022, characterized by a 98% rise in densely populated metropolitan areas as well as a 120% increase in rural built-up areas, exceeds the pace observed in many comparable cities

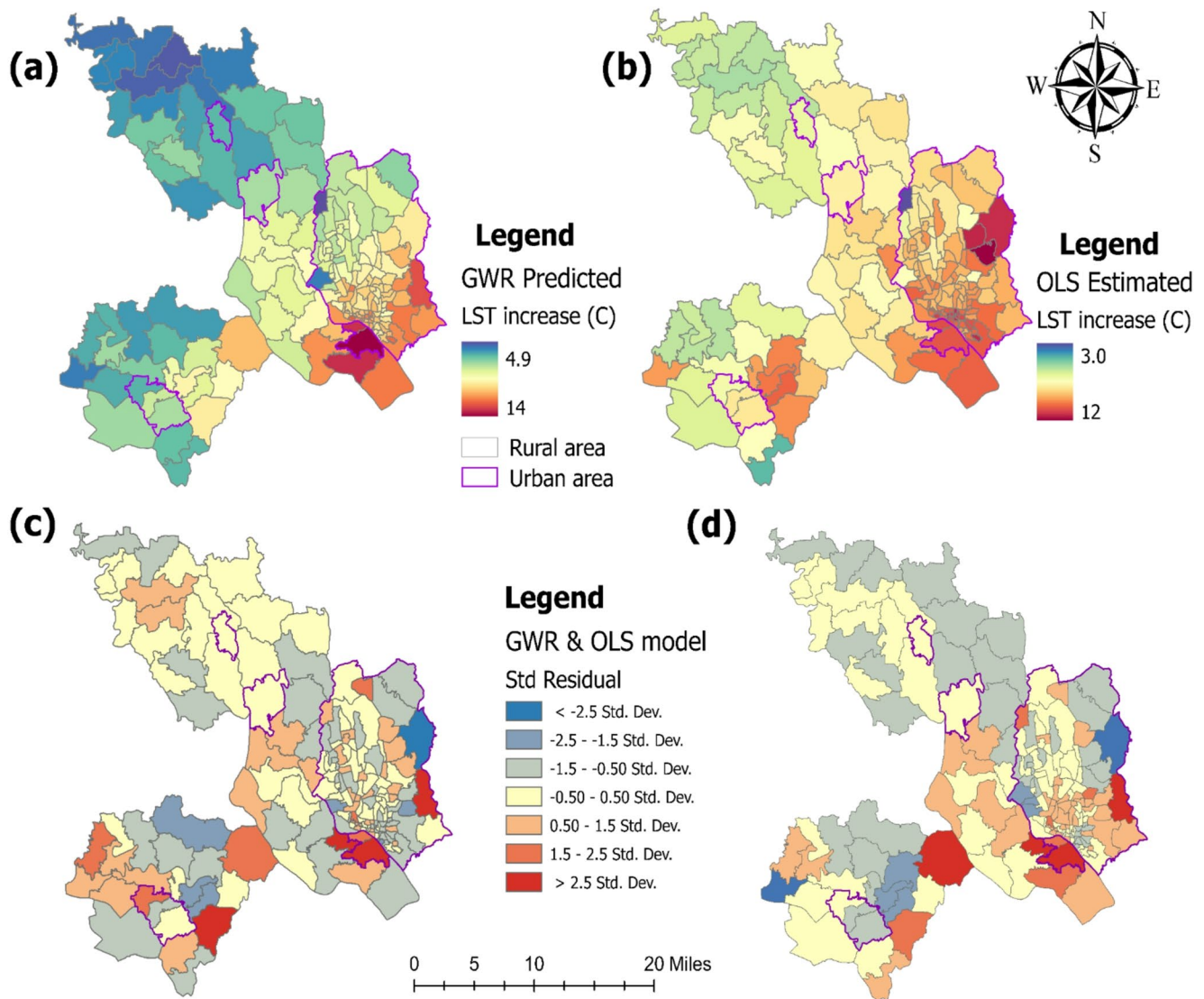


Fig. 17 Comparison of LST predictions and standard residuals from GWR and OLS models: (a) GWR predicted LST, (b) OLS predicted LST, (c) GWR standard residuals, and (d) OLS standard residuals

globally. (Hassan and Southworth 2017) found that Dhaka's built-up area increased by 88.78% between 1990 and 2011, highlighting the continuation of this trend in our study period. This pace of growth is consistent with the results from Xu et al. 2019; who noted that South Asian cities, including Dhaka, experienced some of the highest rates of urban growth worldwide, with significant implications for sustainability and climate resilience.

The loss of 48% of urban vegetation and 19% of water bodies in Dhaka presents a critical challenge for environmental sustainability. Ishtiaque et al. 2017 reported a similar trend, noting a 34% decrease in vegetation cover in Dhaka between 1989 and 2014, emphasizing the ongoing nature of this issue. This rapid loss of green spaces and water bodies is particularly concerning given their role in mitigating urban heat and providing ecosystem services. In contrast,

Swapan et al. 2017 found that some Asian cities have successfully implemented urban greening strategies, increasing their vegetation cover by up to 20% over a similar period, highlighting potential pathways for Dhaka to enhance its urban ecology.

The transformation of rural areas into suburban regions to built-up surfaces signals a shift in the economic base of these regions. Ahmed et al. 2019 found that agricultural land in Dhaka's fringe areas decreased by 37% between 1990 and 2015, aligning with our observations of rapid peri-urban development. This trend is further corroborated by Hou et al. 2020; who found that Dhaka's peri-urban agricultural land loss rate was nearly twice that of other South Asian megacities, emphasizing the need for targeted peri-urban land management strategies to address food security concerns and preserve ecosystem services.

4.2 Demographic shifts and their implications

The population dynamics revealed in our study illustrate Dhaka's struggle to accommodate rapid growth within its existing urban fabric. Faisal et al. 2021 noted that Dhaka's population density increased by 35% between 2001 and 2011, corroborating our findings of continued densification. The intensification of population density in the urban core, particularly in Dhaka City Corporation (DCC), where density increased to over 31,320 people in the study location, exemplifies the challenges of high-density urban living in the Global South. This density surpasses that of many other megacities, as noted by Kabisch et al. 2016 in their comparative study of city population densities worldwide.

The emerging patterns of suburbanization, evident in the growth of peri-urban areas like Savar and Keraniganj, signal a new phase in Dhaka's urban development. (Kontgis et al. 2014) documented similar patterns in other rapidly growing Asian cities, where uncontrolled peri-urban growth has led to significant environmental degradation and infrastructure challenges. Building on this, Dewan et al. 2021 noted that the built-up area in Dhaka's peri-urban regions increased by 136% between 1990 and 2019, reflecting the city's outward expansion and the associated challenges of managing urban sprawl.

4.3 Thermal environment evolution and UHI effect

Thermal and LULC changes observed in Dhaka from 1998 to 2022 align with established patterns in humid tropical regions, where urban expansion elevates LST, while vegetation and water bodies exert cooling effects (Oke 1982; Ziter et al. 2019). These localized land use impacts do not contradict global warming trends but rather operate concurrently. In fact, the recorded LST increase of approximately 0.98 °C and the urban–rural summer temperature differential reaching up to 6 °C exceed global average warming rates (IPCC 2023), underscoring the potent influence of rapid urbanization in megacities like Dhaka. Thermal dynamics in Dhaka differ markedly from those in arid or semi-arid climates due to variations in albedo, evapotranspiration, and soil moisture retention (Arnfield 2003). While the observed patterns are consistent with humid tropical conditions, they remain significant in demonstrating how land use change can locally amplify warming beyond global climate trends.

A particularly strong correlation between NDBI and LST further confirms the role of developed surfaces in intensifying the UHI effect. For instance, Yang et al. 2017 reported a 0.9 °C increase in LST for every 0.1 rise in NDBI in Beijing, whereas our study found a 1.5 °C increase for the

same NDBI change in Dhaka—highlighting the severity of urban heating in this context. Rahman et al. 2020 similarly observed that areas with high NDBI values in Dhaka experienced LSTs up to 4 °C higher than less developed zones.

Conversely, NDVI exhibited a clear cooling effect, reinforcing the importance of urban vegetation in mitigating heat stress. Our analysis revealed an average LST reduction of 0.8 °C in urban areas and 0.5 °C in rural areas for every 0.1 increase in NDVI. These findings are consistent with Bellón et al. (2017), who reported that vegetated zones in Dhaka were 2–3 °C cooler than built-up areas. This evidence supports the role of green infrastructure as a critical climate adaptation strategy, echoing conclusions from Ziter et al. (2019) regarding the cooling benefits of urban tree canopy cover. However, it is important to note that the thermal impacts of LULC change are highly context-dependent. UHI intensity and land cover–temperature relationships vary with local climate, urban morphology, and socioeconomic conditions (Li and Zhou 2019; Oke 1982). Therefore, while the findings presented here are robust within Dhaka's humid tropical setting, they underscore the need for regionally tailored approaches to urban climate adaptation.

4.4 Spatial non-stationarity and policy implications

Our spatial modeling results, comparing OLS and GWR approaches, reveal the spatial non-stationarity of land cover–LST relationships in Dhaka. The superior performance of the GWR model (R^2 increasing from 0.578 to 0.9005) highlights the importance of considering local contexts in urban climate studies. Wang et al. 2016 similarly found significant spatial variations in the connection between land cover types and LST in their study of Chinese cities. This observation is reliable with recent studies on Dhaka's urban climate, such as (Dewan et al. 2021), who found significant spatial variations in the relationship between land use and surface temperature across the city.

The varying strengths of connection within land cover indices and LST across different parts of the city suggest that one-size-fits-all approaches to urban heat mitigation are likely to be ineffective. Santamouris et al. 2015 emphasized the importance of tailored strategies that consider local urban morphology and climate conditions for urban overheating and heat island mitigation. Building on this, Sharmin et al. 2017 highlighted the importance of tailored interventions that consider local urban morphology and climate conditions specifically for Dhaka's urban cooling strategies. This spatial heterogeneity underscores the need for localized, context-specific interventions rather than blanket policies, providing a foundation for more effective Urban development and climate resilience strategies in Dhaka and similar rapidly urbanizing cities.

4.5 Limitations, challenges, and implications for future urban sustainability

While this study offers valuable insights into the spatio-temporal dynamics of LULC and LST in Dhaka, several limitations must be acknowledged to guide future research. A primary constraint lies in the reliance on decadal satellite imagery, which limits the ability to capture short-term or seasonal land use changes that may significantly influence surface temperature. Consequently, certain urban developments or climate-related events occurring between observation years may have gone undetected. Another challenge concerns data consistency. Although Landsat provides a robust long-term dataset, transitions between sensor generations (e.g., from TM to OLI/TIRS) may introduce subtle discrepancies, despite the application of standard pre-processing techniques. Additionally, while satellite-derived LST offers valuable spatial coverage, it does not always align with near-surface air temperatures experienced by residents. The absence of extensive ground-based validation may therefore affect the precision of thermal estimates. Moreover, the study does not incorporate socioeconomic or infrastructural variables—such as income distribution, building height, surface albedo, or anthropogenic heat emissions—which can significantly influence urban thermal patterns. Integrating these factors in future analyses would enhance the explanatory power of LST models and support more targeted policy interventions. Despite these limitations, the findings have critical implications for sustainable urban development in Dhaka. The rapid decline of green spaces and water bodies, coupled with intensifying UHI effects, underscores the urgent need to embed environmental considerations into urban planning frameworks. Strategic priorities should include investment in green infrastructure, integrated water management, heat-resilient urban design, and compact, mixed-use development in peri-urban zones (Moniruzzaman et al. 2020; Smith et al. 2021). Strengthening early warning systems for heat stress and enhancing urban climate monitoring are also essential.

Future research should investigate the socioeconomic drivers of land cover change, particularly in rapidly urbanizing fringe areas, and assess the public health impacts of rising urban temperatures—especially among vulnerable populations. Developing and testing context-specific, nature-based solutions that address heat, flooding, and air quality simultaneously could offer scalable models for climate-resilient urbanism. The patterns of land transformation and thermal intensification observed in Dhaka present both challenges and opportunities, offering a foundation for innovative, equitable, and sustainable urban strategies applicable to other rapidly growing cities in the Global South.

5 Conclusion

This study reveals a strong and spatially variable relationship between land cover transformations and seasonal thermal dynamics in Dhaka, one of South Asia's fastest-growing megacities. The findings highlight the complex interplay between rapid urban expansion and intensifying UHI effects. Over the past three decades, Dhaka has undergone substantial landscape changes, marked by a significant increase in built-up areas and a corresponding decline in vegetation (48%) and water bodies (19%). These shifts are closely associated with elevated LST, particularly in developed and barren zones, while vegetated and water-rich areas consistently exhibit cooler thermal profiles.

Rural zones, although experiencing less vegetation loss and lower average temperatures than urban cores, are increasingly subject to development pressures. Between 1998 and 2022, the study recorded an average temperature increase of 0.98 °C, driven by both anthropogenic land use changes and broader environmental factors. Spatial modeling revealed considerable heterogeneity in the relationship between LULC and LST, with geographically weighted regression outperforming global models ($R^2 = 0.901$ vs. 0.578), underscoring the importance of accounting for spatial variation in urban climate research. Notably, summer temperature differences between urban and rural areas reached up to 6 °C, illustrating the localized impact of urbanization on thermal environments—often exceeding the influence of global climate trends.

These results underscore the urgent need to preserve and expand green infrastructure—particularly vegetation and water bodies—as a core strategy for mitigating urban heat and enhancing climate resilience. The study advocates for immediate integration of sustainable development principles into urban planning, including heat-responsive design, water conservation, and compact, mixed-use development in peri-urban areas. The insights generated are directly applicable to urban planners, policymakers, and environmental managers working in rapidly urbanizing contexts.

Furthermore, the methodological framework employed in this study offers a replicable model for analyzing urban climate dynamics in other megacities, particularly within tropical and subtropical regions. By advancing the understanding of how human-driven land use changes influence local climate systems, this research contributes meaningfully to the fields of urban climatology and eco-climatology. It supports evidence-based decision-making for climate adaptation and sustainable urban development across the Global South.

Acknowledgements The authors extend their appreciation to the Ongoing Research Funding program (ORF-2025-848), King Saud University, Riyadh, Saudi Arabia.

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Funding This research work was supported by the Ongoing Research Funding program (ORF-2025-848), King Saud University, Riyadh, Saudi Arabia.

Data availability No datasets were generated or analysed during the current study.

Code availability Not applicable.

Declarations

Ethical approval Not applicable.

Competing interests The authors declare no competing interests.

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